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Long Semester Project



**Politecnico
di Torino**

Topic: Two-stage speed increaser for a wind power plant

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Introduction

Objectives

The aim of this first part of the work is to accomplish the static calculation of shaft 1 and shaft 2 of a speed increaser for a Wind Power Plant. Our gearbox is powered by an airfoil rotor moved by the wind. For our case of study, it is a two-stage speed increaser. The airfoil drives the high-speed that goes to the generator to produce electricity. The input power goes from the planetary gear (rotor side) to shaft 2 (in blue) by means of spur gears at the connection of the sun with shaft 2. The second stage of the speed increment is a wheel-pinion stage, where P1 is the pinion on the shaft one, and W1 is the wheel on shaft 2. Shaft 2 is hollow for the reason of passing the electric cables needed to control the inclination of the blades. We neglect the presence of the auxiliary shaft, and we suppose the whole power is transmitted to shaft 1. The scheme of the system is presented in figure 1.1.

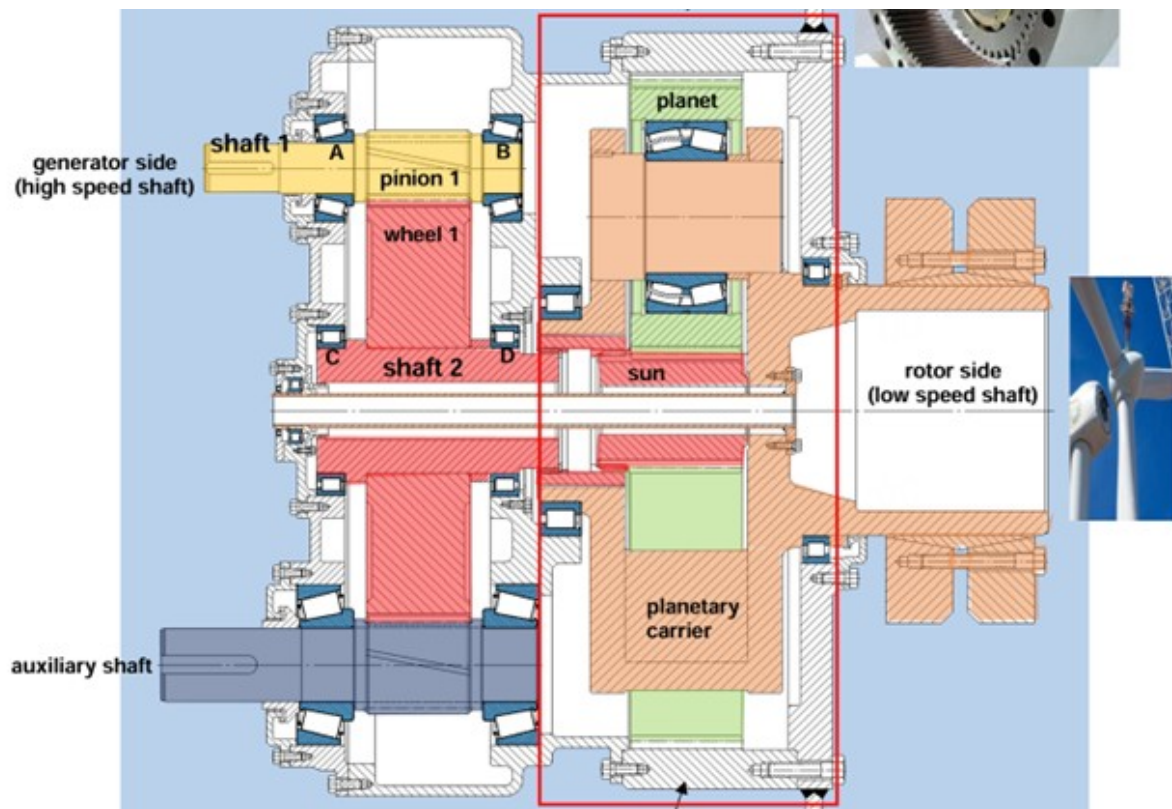


Figure 1: Overall scheme of the gearbox

Data

Herein are presented some data which will be used during the first part of the evaluation.

Table 1. Given data	
Dimensions	1730 × 274800 × 2100 mm
Operating data	
Input power (P)	1150 kW
Input speed (n)	24.5 rpm
Gear data	
Total gear transmission ratio (i)	41.63
Pressure angle (α)	20°
Gears (second stage)	
Helical teeth spiral angle (β)	15°
Bearings and bore diameter (d)	
Bearings A and B - series 32318	d= 90 mm
Bearings C and D - series NCF 2952 V	260mm
Suggested material for the shafts	
Quenched and tempered steel 36NiCrMo16 UNI EN10083	$R_m = 1250 \text{ MPa}$, $R_{eH} = 1050 \text{ MPa}$,

Figure 2: General data

Chapter 1

Part 1 - Design against static failure

1.1 Preliminary calculations

As requested by the subject, we first want to design our system against static failure. Consequently, we will first need some preliminary calculations to set the problem's data.

1.1.1 Estimation of dimensions

The subject does not provide the exact dimensions of the system. However, we know the exact dimensions of the bearings. Especially the bore diameter of the four bearings. Measuring the dimensions of the bearings on our drawing and knowing their real dimensions, we can compute the scale of our drawing. Thus, we will be able to measure and estimate all the dimensions we need to perform the calculations. The scale factor is:

$$\text{Scale Factor} = \frac{\text{Measurement}}{\text{Real Dimension}} \quad (2.1)$$

You can here after see the values measured in correspondence with the real dimensions for the bore diameter and the scale factor for the bearings A and B:

	Bearing A	Bearing B
Measured [mm]	12	27
Real dimensions [mm]	90	260
Scale Factor	7.5	9.6

As we measure all the dimensions and estimate the real dimensions with the scale factor, we found values with a lot of digits. However, the measuring precision is quite low and the values doesn't correspond to the standards of engineering values. Consequently, we decide to round the values and approximate the dimensions to be more consistent. We propose hereafter the dimensions of both shafts.

1.1.2 Torque and Gear forces

In the shaft 1, the torque is applied on the pinion. The keyseat is used to transmit the torque to the generator.

A further important step is to choose the role of each bearing from the scheme. With this purpose each shaft is considered as a statically determined structure. We had to choose the direction of rotation of the whole system: looking from the z positive direction, the pinion is rotating counterclockwise, while the wheel clockwise. Shaft 1 has bearing A and bearing B that can support both, axial and radial forces. But due to the choice of angular speed direction, the axial force will act mostly on bearing A, while still a portion of that force is handled by bearing B. This action of the forces will make bearing A to be locating and bearing B non-locating. Regarding shaft 2, bearing D which behave as an hinge (can support both radial and axial forces) although it is a roller because it is a locating bearing, and bearing C is a roller(support only radial forces).

At this stage we can evaluate the input torque (T_1) applied on the shaft 1 and torque transmitted to shaft 2 (T_2) as following:

$$T_{in} = \frac{P \cdot 60 \cdot 1000}{2\pi n} = 448.3kNm \quad T_{out} = \frac{T_{in}}{i} = 107.7kNm \quad (2.3)$$

Where P is the input power and i (overall transmission ratio). Furthermore, it is possible to evaluate the forces exchanged between the gears on shaft 1 and shaft 2. The forces transmitted by the wheel on pinion P can be evaluated as:

The forces transmitted by the pinion P can be evaluated as:

$$F_{t1} = \frac{T_{out}}{r_1} = 191.4kN$$

in which r_1 is the primitive radius of pinion 1.

$$F_{a1} = F_{t1} \cdot \tan \beta = 51.3kN$$

$$F_{r1} = F_{t1} \cdot \frac{\tan \alpha}{\cos(\beta)} = 72.1kN$$

considering that α is the pressure angle and β is the helix angle.

While for the shaft 2 we can highlight that:

$$F_{a2} = F_{a1} \quad F_{r2} = F_{r1}$$

Instead, we can evaluate also torque on the shaft 2 T_2

$$T_2 = F_{t2} \cdot r_2 = 86.2kNm$$

where r_2 is the primitive radius of the wheel 2

1.2 Shaft 1

For the evaluation of reaction forces we need to establish a reference system and some points showed in the next sketches. Also we need to perform the evaluation of the forces in both plane x-y and x-z to have a complete frame of our shaft. Using the sketches we can perform the equation of equilibrium on shaft 1.

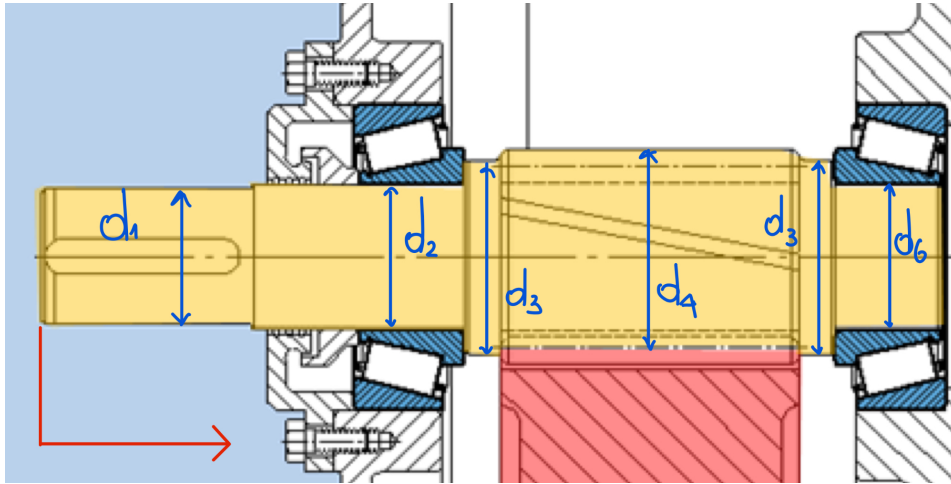
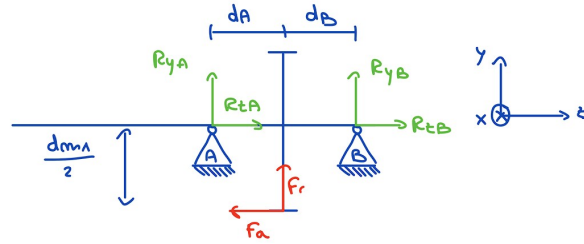


Figure 1.1: Shaft 1

Diameters	Value [mm]
d1	82.5
d2	90
d3	120
d4	112.5
d3	120
d6	90

1.2.1 Reaction Forces

Considering always counterclockwise rotation: **y-z plane**

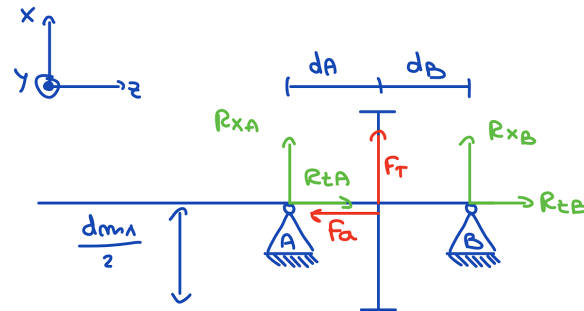


$$\rightarrow) R_{zA} + R_{tzB} = F_a$$

$$\uparrow) R_{yA} + R_{yB} = -F_r$$

$$A) F_r \cdot d_A - F_a \cdot \frac{dm}{2} + R_{yB}(d_A + d_B) = 0$$

x-z plane



$$\rightarrow) R_{zA} + R_{tzB} = F_a$$

$$\uparrow) R_{xA} + R_{xB} = -F_T$$

$$A) F_T \cdot d_A + R_{xB}(d_A + d_B) = 0$$

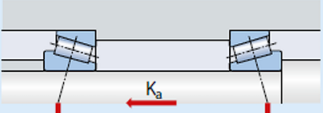
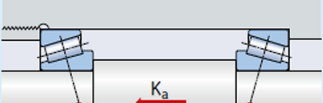
Using those equations and using the needed values the results of the reaction forces on shaft 1 will be:

1.2.2 Evaluation and Plots of Geometrical Features

To have a better idea of the variation of the various geometrical interesting quantity of the system, is better to plot how the area (A), bending inertia (M_b) and torsion inertia (M_t) changes along the x coordinate of the shaft.

Reaction Loads (Shaft 1)		
x-z plane		
RxB	-95707.1105	N
RxA	-95707.1105	N
y-z plane		
RyB	-26422.5257	N
RyA	-45704.2122	N
Axial Reactions (Case 2b from catalogue)		
RzB	29202.1958	N
RzA	80491.4818	N

In order to evaluate R_{zA} and R_{zB} I must look at the tables for the readings:

Back-to-back 	Case 2a $\frac{F_{rA}}{Y_A} \leq \frac{F_{rB}}{Y_B}$ $F_{aA} = F_{aB} + K_a$ $F_{aB} = \frac{0,5 F_{rB}}{Y_B}$ $K_a \geq 0$
Face-to-face 	Case 2b $\frac{F_{rA}}{Y_A} > \frac{F_{rB}}{Y_B}$ $F_{aA} = F_{aB} + K_a$ $F_{aB} = \frac{0,5 F_{rB}}{Y_B}$ $K_a \geq 0,5 \left(\frac{F_{rB}}{Y_B} - \frac{F_{rA}}{Y_A} \right)$
	Case 2c $\frac{F_{rA}}{Y_A} > \frac{F_{rB}}{Y_B}$ $F_{aA} = \frac{0,5 F_{rA}}{Y_A}$ $F_{aB} = F_{aA} - K_a$ $K_a < 0,5 \left(\frac{F_{rA}}{Y_A} - \frac{F_{rB}}{Y_B} \right)$

We must compare $\frac{F_{tA}}{Y_A} = \frac{\sqrt{R_x^2 + R_y^2}}{Y_A}$ with $\frac{F_{rB}}{Y_B} = \frac{\sqrt{R_x^2 + R_y^2}}{Y_B}$ and then calculate $K_a = F_a$.

To know in which case we are operating (2a, 2b, 2c) and then calculate R_{zA} and R_{zB} .

To be brief we will show the formulas used to get the results and then we only report the charts in the various cases. The geometrical features are evaluated as follows:

$$A = \frac{\pi \cdot d^2}{4} \quad W_b = \frac{\pi \cdot d^3}{32} \quad W_t = \frac{\pi \cdot d^3}{16} \quad (2.9)$$

Below are shown the graphs of trend of area (A), bending inertia (M_b) and torsion inertia (M_t) along shaft 1.

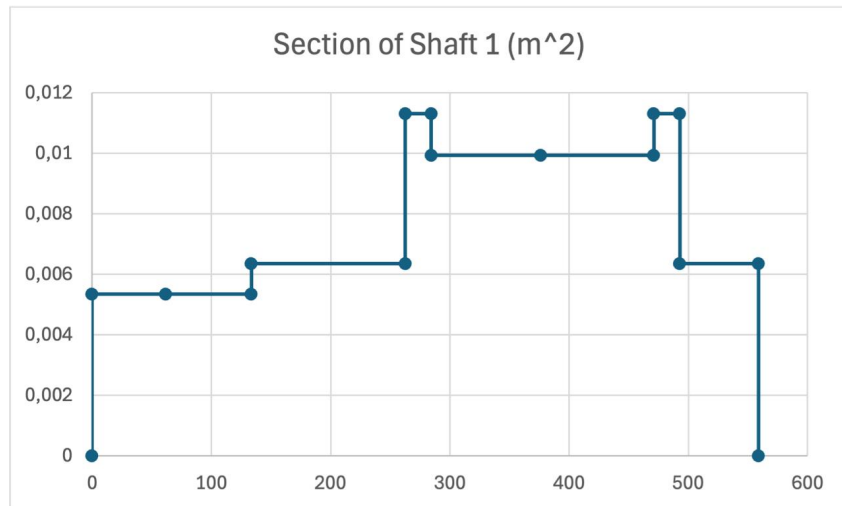
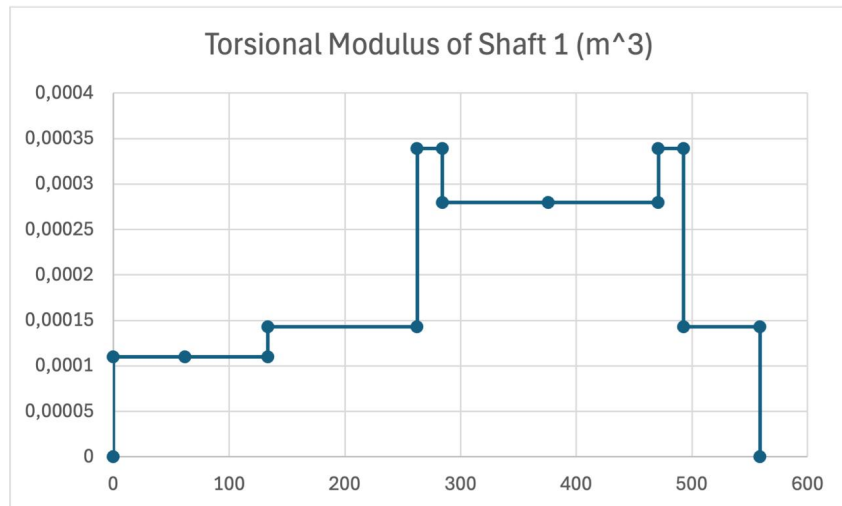
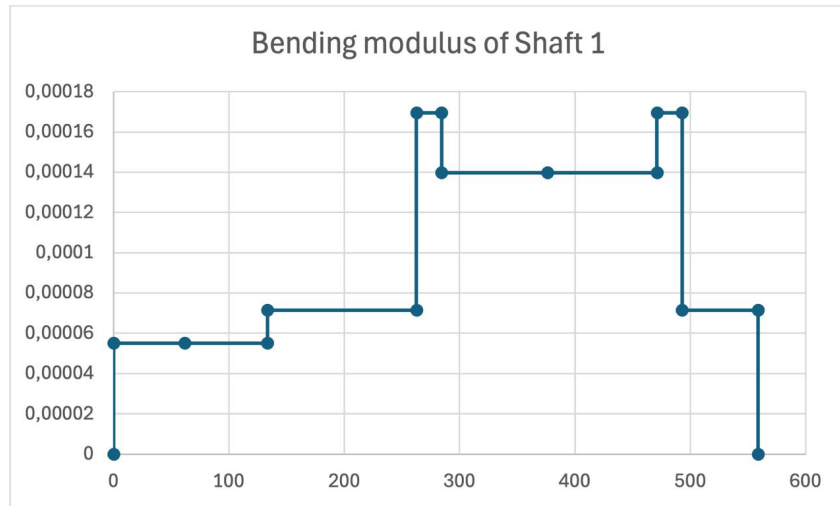


Figure 1.2: Section Area of shaft 1 along z



1.2.3 Evaluation and Plots of Loads

In order to evaluate the variation of axial load (or normal), bending moment and torsional moment along the shaft 1 we must carry out a section by section analysis. To do so we've used the reference model here showed.

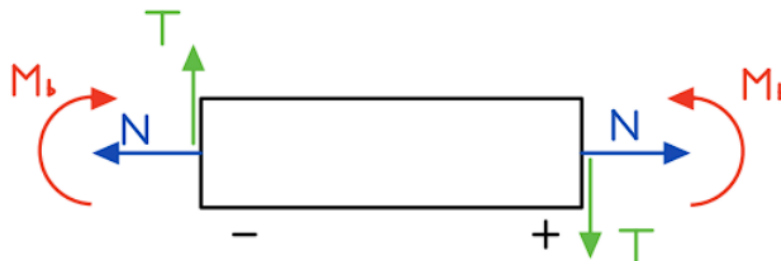
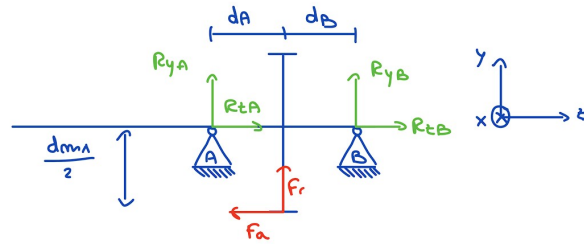


Figure 1.3: Convention of signs

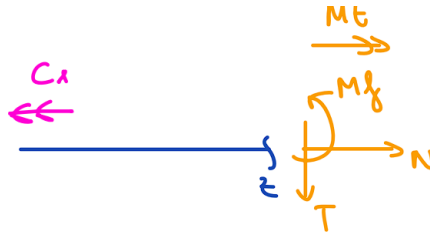
To perform the evaluation we will show the evaluation of Stress characteristics in the x-y plane, and then for the x-z plane, as well as shaft 2 we will only show our result for briefness reasons.

y-z plane

Using the same sketch of reaction forces we can start to evaluate the stresses with the reference frame z which begins from the right to the left of the shaft as showed in figure.



$$0 < z < z_A:$$

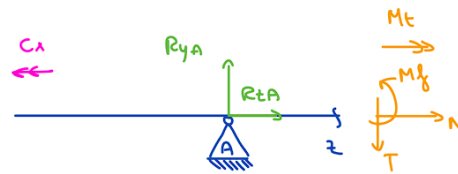


$$N = 0 \text{ N}$$

$$T = 0 \text{ N}$$

$$M_{tz} = C_1 \text{ N m}$$

$z_A < z < z_G$ where z_G is the coordinate of the gear :



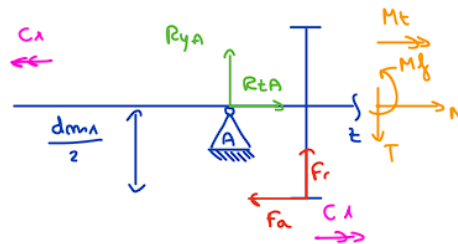
$$N = -R_{zA} \text{ N}$$

$$T = R_{yA} \text{ N}$$

$$M_{fx} = R_{yA} \cdot z \text{ N m}$$

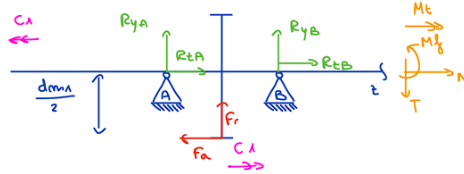
$$M_{tz} = C_1$$

$z_G < z < z_B$:



$$\begin{aligned}
N &= F_a - R_{za} \\
T &= F_r + R_{ya} \\
M_{fx} &= R_{ya} \cdot (d_A + z) + F_r \cdot z + F_a \cdot \frac{dm}{2} \\
M_{tz} &= 0
\end{aligned}$$

$z_B < z < end :$



$$\begin{aligned}
N &= 0 \\
T &= R_{yA} + F_r \\
M_{fx} &= 0 \cdot x \\
M_t &= 0
\end{aligned}$$

The result can be gathered in diagrams which are below reported.

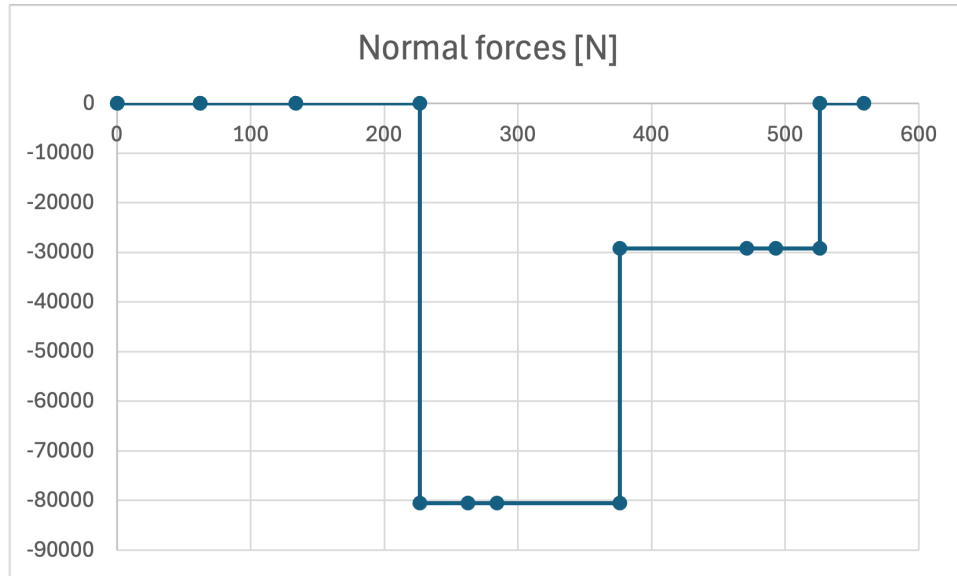


Figure 1.4: Axial Load

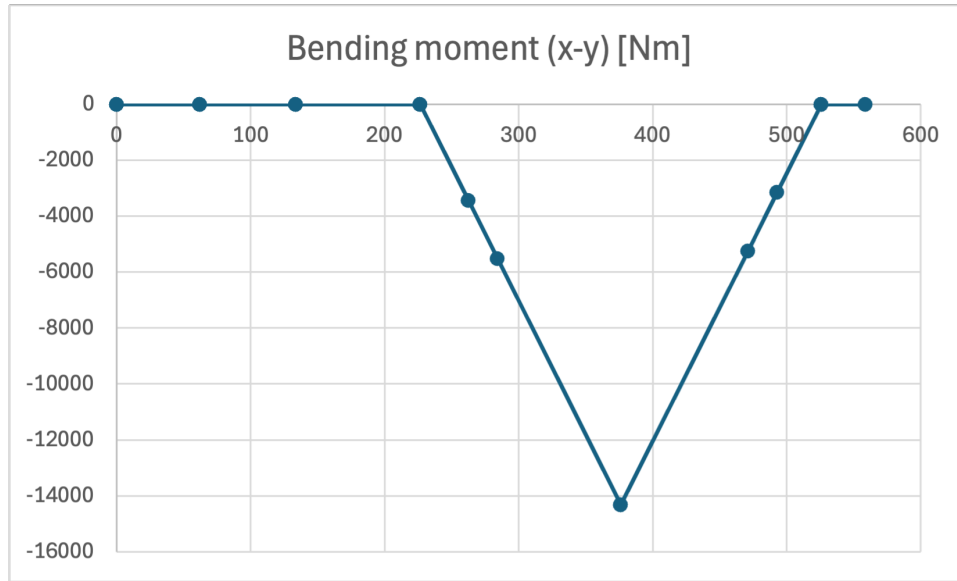
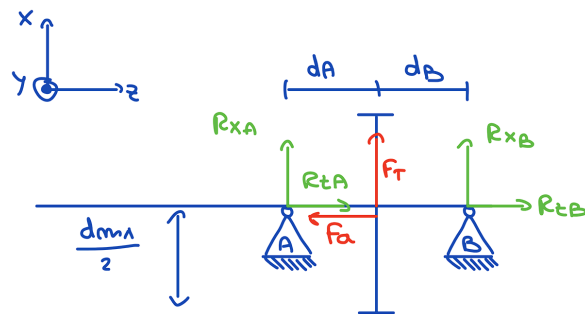


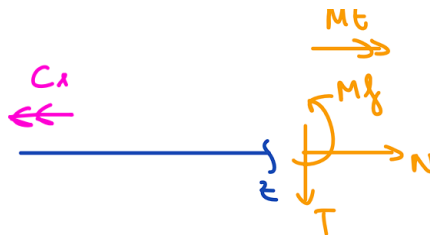
Figure 1.5: Bending moment along SHAFT 1 on Y-Z plane

x-z plane

Using the same sketch of reaction forces we can start to evaluate the stresses with the reference frame z which begins from the right to the left of the shaft as showed in figure.

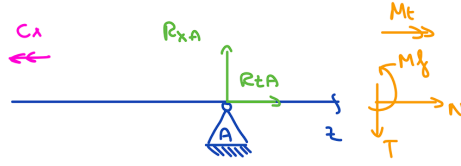


$$0 < z < z_A:$$



$$\begin{aligned} N &= 0 \text{ N} \\ T &= 0 \text{ N} \\ M_{tz} &= C_1 \text{ Nm} \\ M_{fy} &= 0 \text{ Nm} \end{aligned}$$

$z_A < z < z_G$ where z_G is the coordinate of the gear :



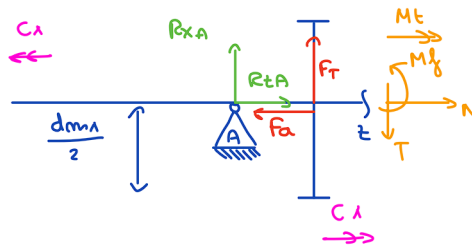
$$N = -R_{zA} N$$

$$T = R_{yA} N$$

$$M_{fy} = R_{xA} \cdot z N m$$

$$M_{tz} = C_1$$

$z_G < z < z_B :$



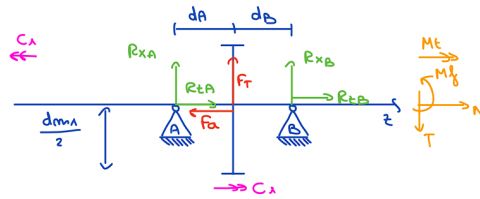
$$N = F_a - R_{za}$$

$$T = F_T + R_{xa}$$

$$M_{fy} = R_{xa} \cdot (d_A + z) + F_T \cdot z$$

$$M_{tz} = 0$$

$z_B < z < end :$



$$N = 0$$

$$T = 0$$

$$M_{fy} = 0$$

$$M_{tz} = 0$$

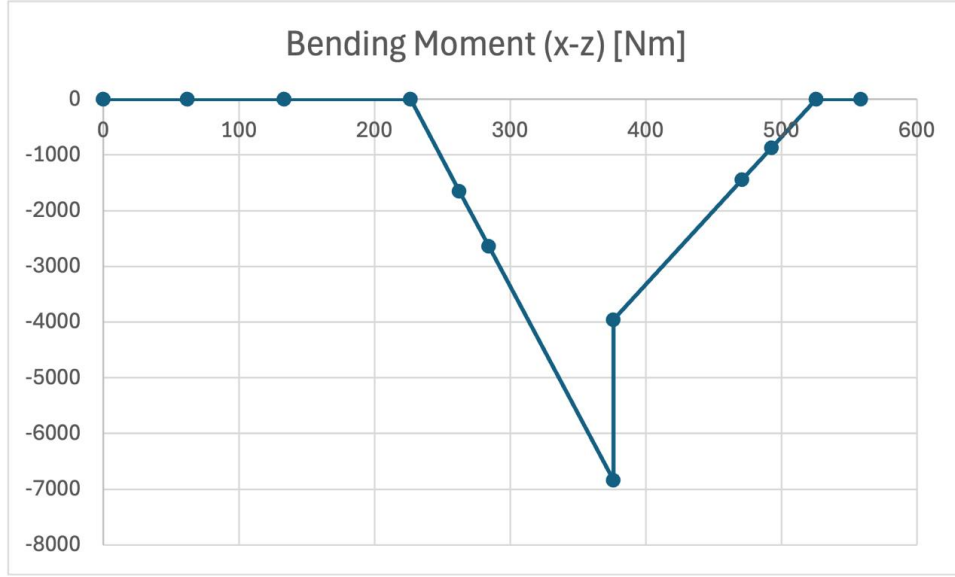


Figure 1.6: Bending moment along SHAFT 1 on X-Z plane

1.2.4 Stress evaluation

Now that the stresses and the changes in cross area are know along all shaft 1, we can proceed evaluating the stresses. Starting with **Normal stress**, defined as:

$$\sigma_n = \frac{N}{A} \quad (2.10)$$

Where the area, section by section, is given by formula 2.9. As it can be seen, compression stresses are acting on shaft 1.

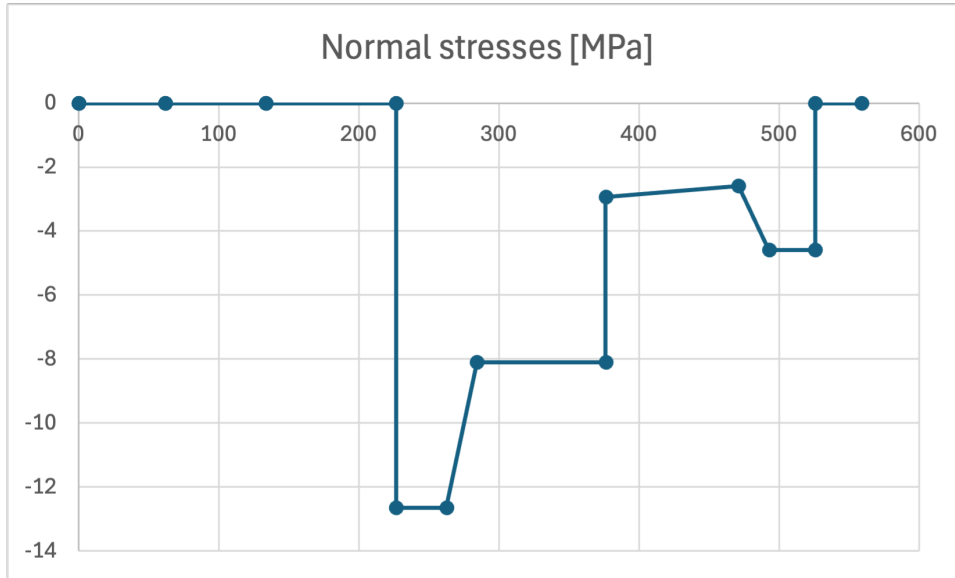


Figure 1.7: Normal stresses

Continuing with the **Bending Stress** along $x - y$ plane, they are defined as:

$$\sigma_{b,xy} = \frac{M_{xy}}{W_b} \quad (2.11)$$

Where M_{xy} is the bending moment along the shaft, whereas W_b is the bending modulus, also given by equation 2.9.

Then, by analyzing the bending moment along $x - z$ plane, we have that:

$$\sigma_{b,xz} = \frac{M_{xz}}{W_b} \quad (2.12)$$

By combining then the two moment M_{xy} and M_{xz} in a global moment, with the following formula:

$$M_{tot} = \sqrt{M_{xy}^2 + M_{xz}^2} \quad (2.13)$$

Then by recalling the 2.9, we will have the following diagrams.

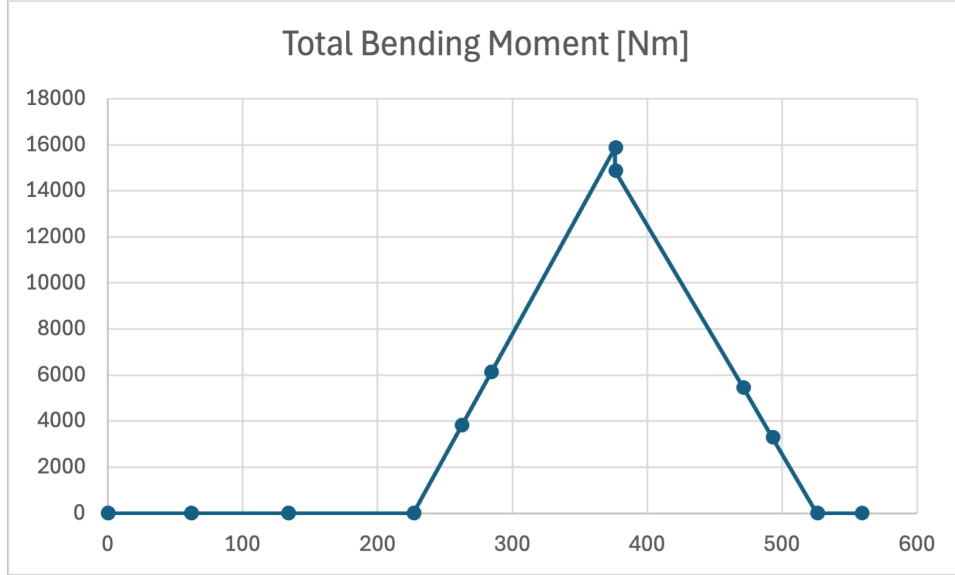


Figure 1.8: Equivalent Total bending moment

Of course, the bending stress jump that was introduced in shaft 1 due to the tangential force on plane $x - y$ (seen by the shaft as a concentrated bending) is therefore visible also in the total bending stress. Finally, by introducing the shear stresses due to torsion, we will have:

$$\tau = \frac{T_1}{W_t} \quad (2.14)$$

Where T_1 is the torque witnessed by shaft 1 (constant along shaft), and W_t is the torsional inertia of the shaft, given by 2.9.

Now, our aim is to compute the equivalent stresses, which will be useful later when computing the safety factor. To compute them, the most conservative hypotheses needs to be used: for this reason, Tresca's criterion is used:

$$\sigma_{eq} = \sqrt{(\sigma_n + \sigma_{b,tot})^2 + 4\tau^2} \quad (2.15)$$

Where σ_n is the normal stress due to normal force N , $\sigma_{b,tot}$ is the total bending stress derived from M_{tot} , and finally τ is the shear stress derived from T_1 . By plotting the result, we get what follows:

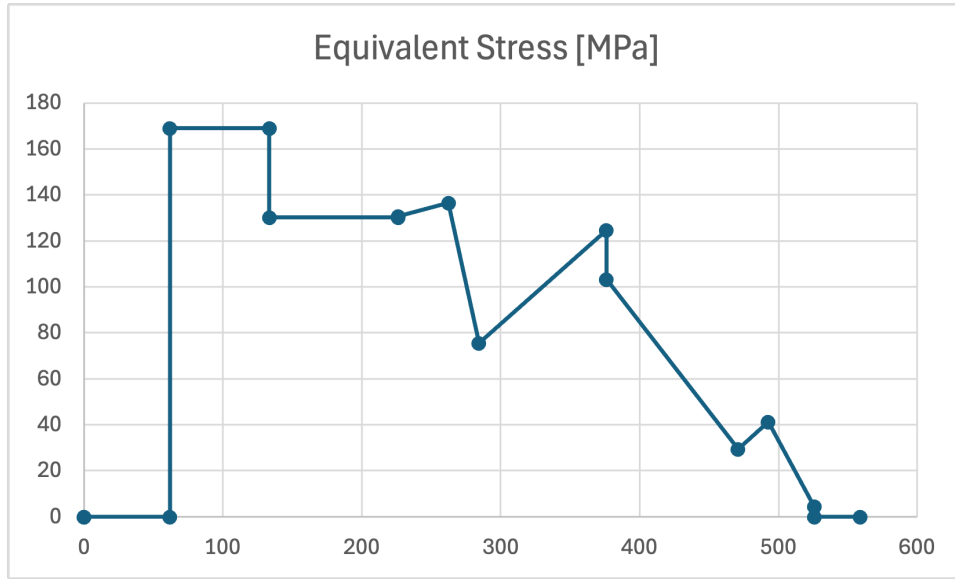


Figure 1.9: Equivalent stress along shaft 1 by using Tresca's criterion

1.2.5 Computation of Safety factor

By the previous calculations, it is possible to obtain the maximum equivalent stress that is witnessed by the shaft, and its location. Precisely, the most excited section (the one with the highest stresses) is the **keyseat**, with a:

$$\sigma_{eq,max} \approx 169 [MPa]$$

The Safety factor is defined considering the Yield Stress of the shaft (without considering any notch effect in this case):

$$SF = \frac{R_{e,N}}{\sigma_{eq,max}} \quad (2.16)$$

Which for the first shaft turns out to be:

$$SF = 6.2$$

1.3 Shaft 2

As we have done before for the evaluation of reaction forces on shaft 1, we need now to establish a reference system on shaft 2 and compute the different reactions at the bearings. Also, the evaluation of the forces is needed both in plane x-z and y-z to have a complete frame of our shaft.

Reaction Loads (Shaft 2)		
x-z plane		
RxD	97210.4	N
RxC	94203.9	N
y-z plane		
RyD	99398.8	N
RyC	-27272	N
Axial Reaction		
RzD	-51289.3	N

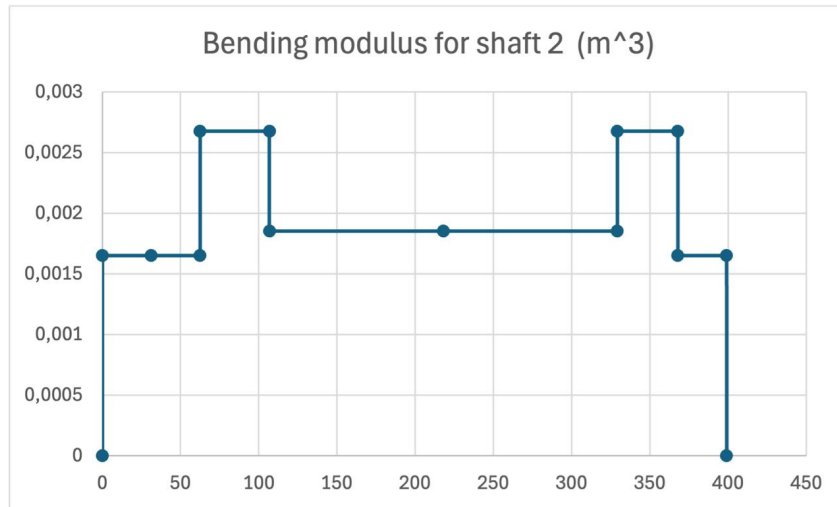
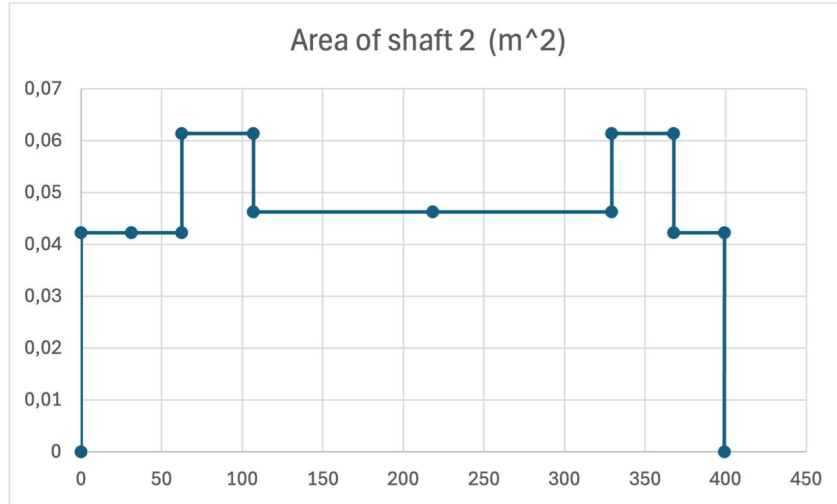
1.3.1 Evaluation and Plots of Geometrical Features

As it has been done for shaft 1, to have a better idea of the variation of the various geometrical interesting quantity of the system, is better to plot the changes of area (A), bending inertia (M_b) and torsion inertia

(M_t) along the shaft. To be essential we will show the formulas used to get the results and then we only report the charts in the various cases. Also for shaft 2 the geometrical features are evaluated as:

$$A = \frac{\pi \cdot d^2}{4} \quad W_b = \frac{\pi \cdot d^3}{32} \quad W_t = \frac{\pi \cdot d^3}{16} \quad (2.17)$$

Below are shown the graphs of trend of area (A), bending inertia (M_b) and torsion inertia (M_t) along shaft 2.



1.3.2 Evaluation and Plots of Loads

The same calculations are performed for shaft 2. We are plotting here only the results as the steps are the same.

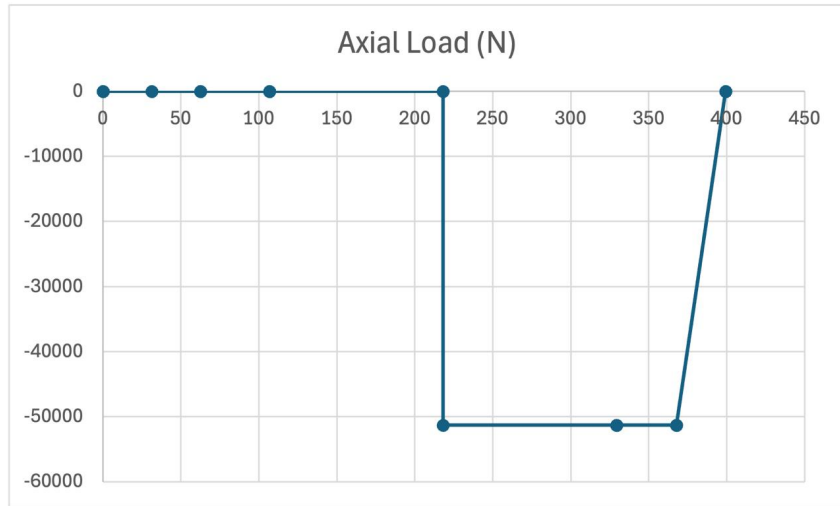


Figure 1.10: Normal force along axial direction for shaft 2

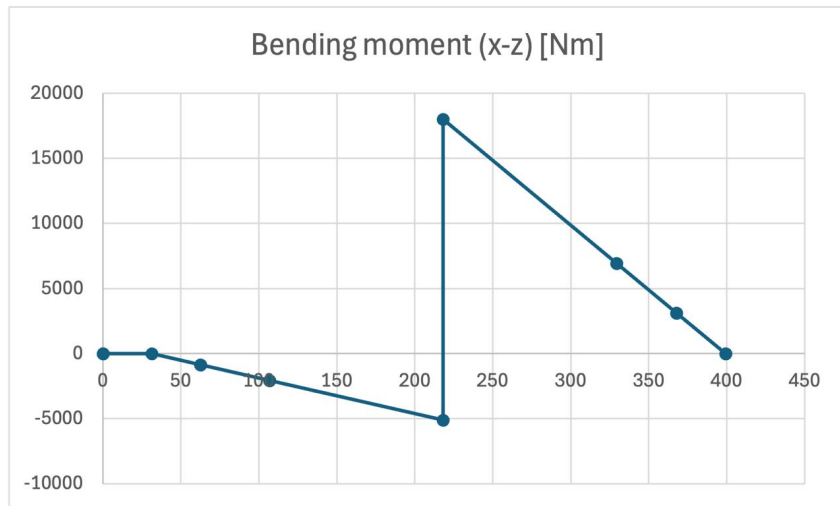


Figure 1.11: Bending moment for shaft 2

1.3.3 Stress evaluation

Now that forces and moments are known section by section, as well as the areas and the different modulus W_t and W_b , we are able to compute the stresses.

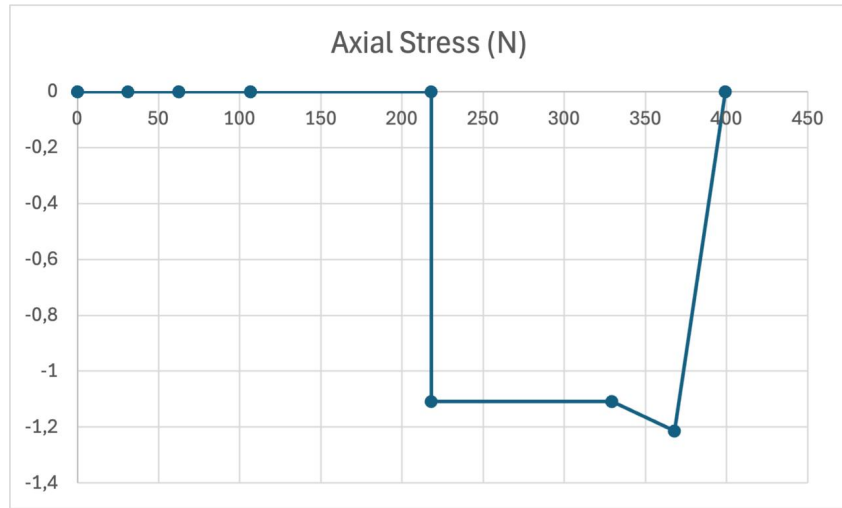


Figure 1.12: Normal stress for shaft 2

We therefore show also the bending stress deriving from the overall bending moment.

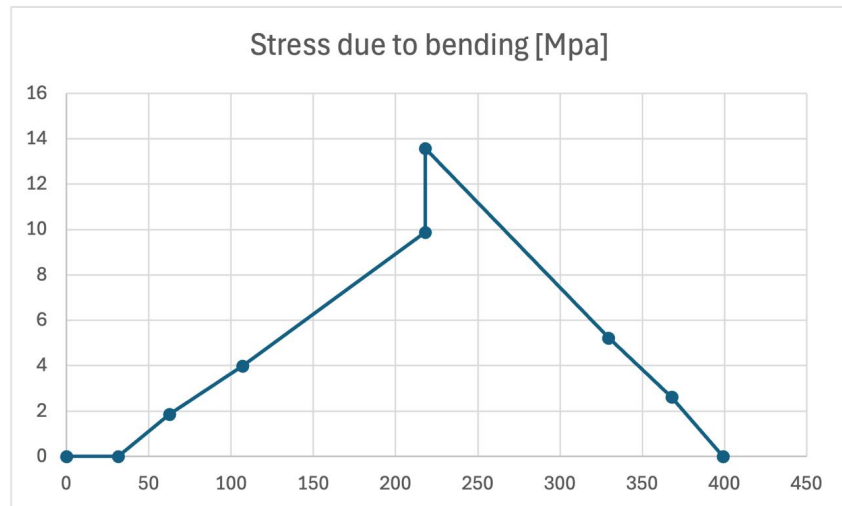


Figure 1.13: Stress due to bending moment for shaft 2

We therefore show also the bending stress deriving from the overall bending moment. Now, by recalling the Tresca criterion for the equivalent stress, we get what follows:

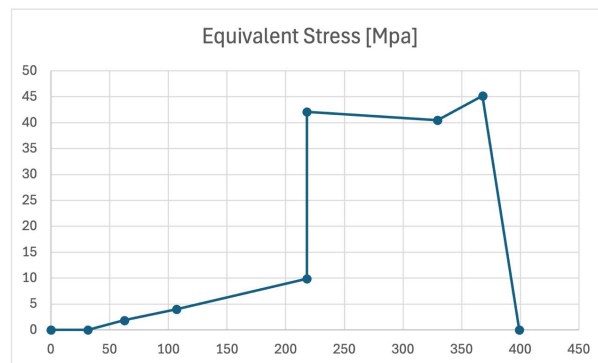


Figure 1.14: Equivalent stress for shaft 2

1.3.4 Computation of Safety factor

Using the graph above, we can detect the value of the maximum stress. In that point, the maximum stress is:

$$\sigma_{eq,max} \approx 45.15 [MPa]$$

The safety factor can then be evaluated, which by recalling formula (2.16):

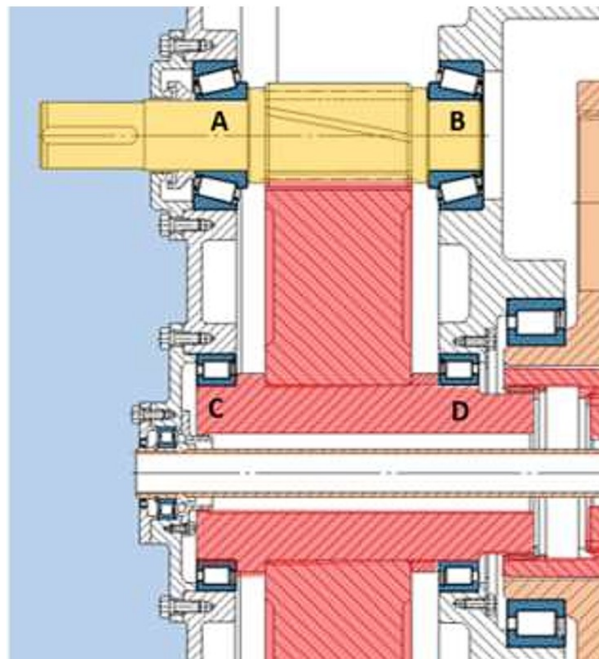
$$SF = 23.25$$

Chapter 2

Part 2 - Design against fatigue failure

2.1 Objective

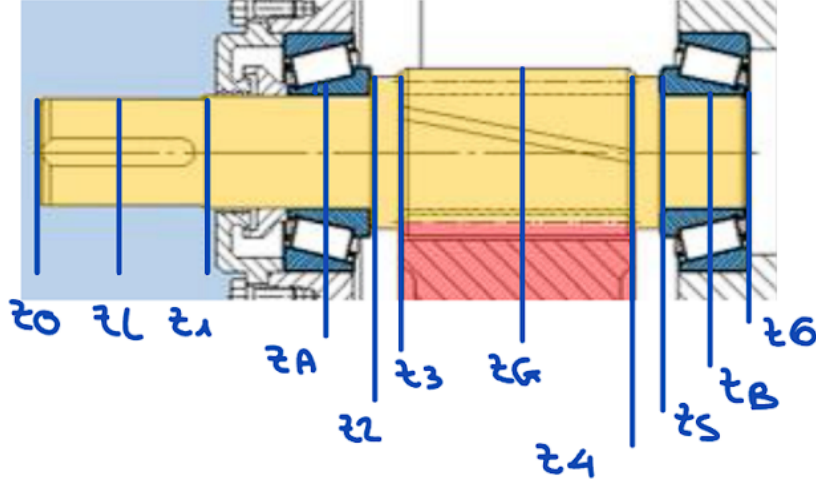
The aim of this second part is to study the speed increaser from a point of view of the fatigue. In particular we are going to prove that the working condition of the shaft 1 and shaft 2 corresponds to infinite life, and in a second moment to use the 'FKM standard' to verify that shaft 1 and shaft 2 can sustain infinite life in some sections. Those sections are showed below to make a clear display.



2.2 Verification to infinite life

Firstly, we will verify that the working condition of shaft 1 and 2 correspond to infinite life condition. This means that the number of cycle is higher than 10^6 cycle.

2.3 Shaft 1



2.3.1 Section z3 - shoulder

In order to verify that the section 1 presenting a shoulder can sustain an infinite life according to the FKM standards, we will allocate the working point in the **Haigh diagram** and verify that this point belongs to the safe regions.

First of all, we compute the R_m of the material.

Shaft 1 is composed of 36NiCrMo16, giving $R_{m,N} = 1250$ MPa.

$$R_m = R_{m,N} \cdot K_A \cdot K_{d,m} \quad (3.1)$$

With $K_A = 0.8$ because we neglect the an-isotropic effect.

We now need to find $K_{d,m}$. With the following parameters:

$$a_{d,m} = 0.3d_{eff} = 112.5d_{eff,N} = 16 \quad (3.2)$$

We can now compute $K_{d,m}$:

$$K_{d,m} = \frac{1 - 0.7686 \cdot a_{d,m} \cdot \log(\frac{d_{eff}}{7.5})}{1 - 0.7686 \cdot a_{d,m} \cdot \log(\frac{d_{eff,N}}{7.5})} = \frac{1 - 0.7686 \cdot 0.3 \cdot \log(\frac{112.5}{7.5})}{1 - 0.7686 \cdot 0.3 \cdot \log(\frac{16}{7.5})} \simeq 0.789 \quad (3.3)$$

Leading to:

$$R_m = 1250 \cdot 0.8 \cdot 0.78 = 788.7 \text{ MPa} \quad (3.4)$$

We compute now the σ_{D-1} : (with $f_w = 0.45$ in FKM standards)

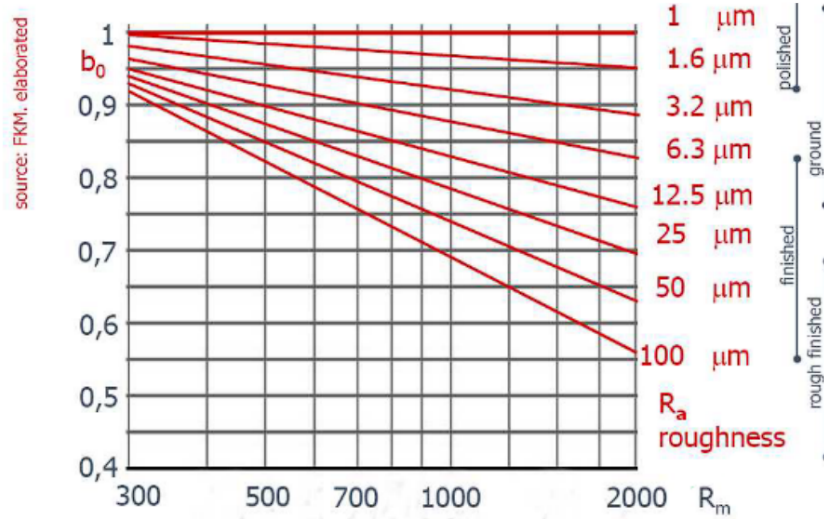
$$\sigma_{D-1} = R_m \cdot f_w = 788.7 \cdot 0.45 = 354.9 \text{ MPa} \quad (3.5)$$

Thanks to the previous computation, we are now able to found the stress $\sigma_{D-1,C}$ applied to our component.

$$\sigma_{D-1,C} = \sigma_{D-1} \cdot K_V \cdot K_S \cdot b_0 \quad (3.6)$$

With $K_V = 1$ and $K_S = 1$ because we neglect the effects of the surface treatment and the coating treatment here.

In order to find the b_0 , parameter corresponding to the surface finishing condition, we will use the following graph.



The section 1 is used to position the bearing B, thus, a high quality of surface roughness is needed and we will consider $R_a = 1.6 \mu m$ corresponding to a polishing finishing.

We finally found $b_0 = 0.96$ and

$$\sigma_{D-1,C} = 354.9 \cdot 1 \cdot 1 \cdot 0.96 \simeq 340.7 \text{ MPa} \quad (3.7)$$

material	steel*	GS	GGG	GT	GG	wrought Al-alloys	cast Al-alloys
a_M	0,35	0,35	0,35	0,35	0	1.0	1.0
b_M	-0.1	0.05	0.08	0.13	0.5	-0.04	0.2

*including stainless steel

Source FKM

For the steel, we obtain $a_M = 0.35$ and $b_M = -0.1$. Giving us: $\alpha = 1.9704$

We want now to plot the Goodman line modified by FKM. Thus, we have to evaluate the equation:

$$\sigma_A = \sigma_{D-1,C} - M_\sigma \cdot \sigma_m \quad (3.8)$$

$$M_\sigma = \frac{\sigma_{D-1,C}}{\alpha \cdot R_m} = a_M \cdot 10^{-3} \cdot R_m + b_M = 0.176 \quad (3.9)$$

We compute now the coordinate of the working point.

First, we choose the steels parameters for the notch effect:

	stainless steel	other steels	GS	GGG	GT	GG	wrought Al-alloys	cast Al-alloys
a_G	0,40	0,50	0,25	0,05	-0,05	-0,05	0,05	-0,05
b_G	2400	2700	2000	3200	3200	3200	850	3200

Figure 3.3: FKM parameters for materials

$$a_G = 0.5 \quad b_G = 2700 \quad (3.11)$$

Based on the diameters shoulder and the type of the shoulder, we were able to pick the radius:

**Table 1 Dimensions for Relief Grooves
(Clause 4)**

All dimensions in millimeters

Form	$r^{a)}$ ± 0.1		t_1 +0.1 0	t_2 +0.05 0	f +0.2 0	g	Recommended co-relation to diameter $d_1^{b)}$ for workpieces	
	Series 1	Series 2					For normal duty conditions ^{c)}	With increased fatigue resistance
E		R 0.2	0.1	-	1	-	Above 1.6 to 3	-
	R 0.4		0.2	-	2	-	Above 3 to 18	-
		R 0.6	0.2	-	2	-	Above 10 to 18	-
		R 0.6	0.3	-	2.5	-	Above 18 to 80	-
	R 0.8		0.3	-	2.5	-	Above 18 to 80	-
		R 1	0.2	-	2.5	-	-	Above 18 to 50
		R 1	0.4	-	4	-	Above 80	-
	R 1.2		0.2	-	2.5	-	-	Above 18 to 50
	R 1.2		0.4	-	4	-	Above 80	-
	R 1.6		0.3	-	4	-	-	Above 50 to 80
F	R 2.5		0.4	-	5	-	-	Above 80 to 125
	R 4		0.5	-	7	-	-	Above 125
		R 0.2	0.1	0.1	1	(0.9)	Above 1.6 to 3	-
	R 0.4		0.2	0.1	2	(1.1)	Above 3 to 18	-
		R 0.6	0.2	0.1	2	(1.4)	Above 10 to 18	-
		R 0.6	0.3	0.2	2.5	(2.1)	Above 18 to 80	-
	R 0.8		0.3	0.2	2.5	(2.3)	Above 18 to 80	-
		R 1	0.2	0.1	2.5	(1.8)	-	Above 18 to 50
		R 1	0.4	0.3	4	(3.2)	Above 80	-
	R 1.2		0.2	0.1	2.5	(2)	-	Above 18 to 50
G	R 1.2		0.4	0.3	4	(3.4)	Above 80	-
	R 1.6		0.3	0.2	4	(3.1)	-	Above 50 to 80
H	R 2.5		0.4	0.3	5	(4.8)	-	Above 80 to 125
	R 4		0.5	0.3	7	(6.4)	-	Above 125
	R 0.4		0.2	0.2	(0.9)	(1.1)	Above 3 to 18	-
H	R 0.8		0.3	0.05	(2.0)	(1.1)	Above 18 to 80	-
	R 1.2		0.3	0.05	(2.4)	(1.5)	-	Above 18 to 50

a) Relief grooves with radii of the series 1 are preferable.
b) The allocation to diameter range is not applicable for short offsets and thin walled parts.
For manufacturing reasons, it would be sensible to effect several relief grooves on a single work piece with different diameters , in the same shape and size .
c) Type G only for work pieces with less load .

Figure 2.1: Radius of the shoulder

Then, with the FKM table we can find $n(r)$ and $n(d)$. The parameters of the section are the following:

$$r = 2.5 \text{ mm} \quad d = 112.5 \text{ mm} \quad (3.12)$$

$\chi_\sigma(r)$	$\chi_\sigma(d)$		$\chi_\sigma(r)$	$\chi_\sigma(d)$																												
$\frac{2}{r}$	$\frac{2}{d}$		$\frac{2.3}{r}$	$\frac{2}{d}$																												
$\frac{2}{r}$	0		$\frac{2.3}{r}$	0																												
$\chi_\tau(r)$	$\chi_\tau(d)$		for : $\chi \leq 0,1 \text{ mm}^{-1}$ $n_{\sigma,\tau} = 1 + \chi_{\sigma,\tau} \cdot 10^{-\left(a_G - 0,5 \cdot \frac{R_m}{b_G}\right)}$ for : $0,1 \text{ mm}^{-1} \leq \chi \leq 1 \text{ mm}^{-1}$ $n_{\sigma,\tau} = 1 + \sqrt{\chi_{\sigma,\tau}} \cdot 10^{-\left(a_G + \frac{R_m}{b_G}\right)}$ for : $1 \text{ mm}^{-1} \leq \chi \leq 100 \text{ mm}^{-1}$ $n_{\sigma,\tau} = 1 + \sqrt[4]{\chi_{\sigma,\tau}} \cdot 10^{-\left(a_G + \frac{R_m}{b_G}\right)}$ WARNING: when $r = 0$, i.e. the notch does not exist, use $\chi(r) = 0$																													
$\frac{1}{r}$	$\frac{2}{d}$																															
$\frac{1.15}{r}$	$\frac{2}{d}$		<table><tr><th></th><th>stainless steel</th><th>other steels</th><th>GS</th><th>GGG</th><th>GT</th><th>GG</th><th>wrought Al-alloys</th><th>cast Al-alloys</th></tr><tr><td>a_G</td><td>0,40</td><td>0,50</td><td>0,25</td><td>0,05</td><td>-0,05</td><td>-0,05</td><td>0,05</td><td>-0,05</td></tr><tr><td>b_G</td><td>2400</td><td>2700</td><td>2000</td><td>3200</td><td>3200</td><td>3200</td><td>850</td><td>3200</td></tr></table>				stainless steel	other steels	GS	GGG	GT	GG	wrought Al-alloys	cast Al-alloys	a_G	0,40	0,50	0,25	0,05	-0,05	-0,05	0,05	-0,05	b_G	2400	2700	2000	3200	3200	3200	850	3200
	stainless steel	other steels	GS	GGG	GT	GG	wrought Al-alloys	cast Al-alloys																								
a_G	0,40	0,50	0,25	0,05	-0,05	-0,05	0,05	-0,05																								
b_G	2400	2700	2000	3200	3200	3200	850	3200																								

By using this table with different configurations of loading conditions now can compute the χ and n_σ :

$\chi_{\sigma b}(r)$	0.92	$n_\sigma(r)$	1.15481
$\chi_{\sigma b}(d)$	0.01778	$n_\sigma(d)$	1.00907
$\chi_{\sigma tc}(r)$	0.92	$n_{\sigma tc}(r)$	1.15481
$\chi_{\sigma tc}(d)$	0	$n_{\sigma tc}(d)$	1
$\chi_\tau(r)$	0.46	$\pi_\tau(r)$	1.10947
$\chi_\tau(d)$	0.01778	$\pi_\tau(d)$	1.00907

Table 2.1: Table of values for various parameters of loading conditions, radius of the notch, and value of diameter d

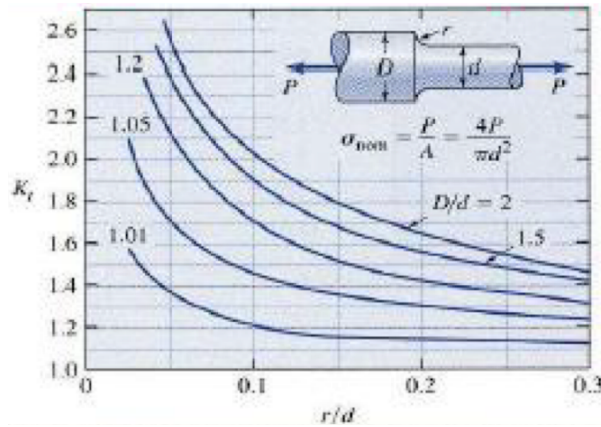


Figure 2.2: Kt for tension/compression case

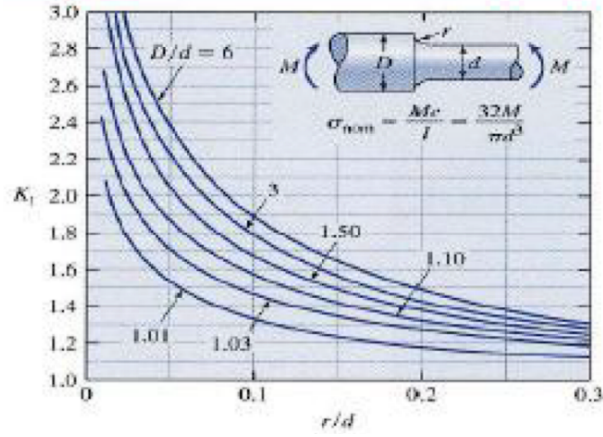


Figure 2.4: Kt for torsion

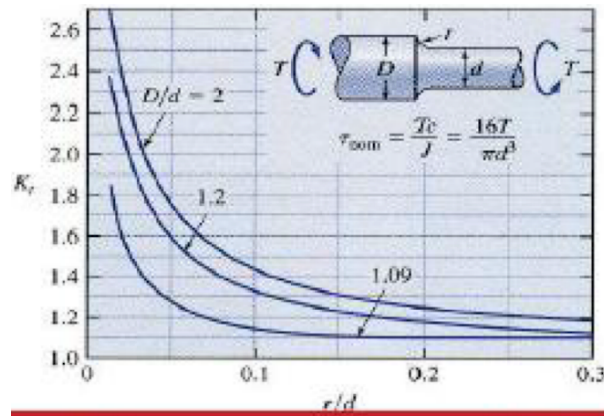


Figure 2.3: Kt for bending

Based on the graphs above, having a ratio r/d equal to 0.0222 and D/d 1.0667 we are able to find K_t for each loading condition:

$K_{t,tc}$	1.93
$K_{t,b}$	2.33
$K_{t,t}$	1.46

Table 2.2: Stress concentration factors

Thus we can use the formula to find K_f ,

$$K_f = \frac{K_t}{n_{\sigma,r} \cdot n_{\sigma,d}}$$

$K_{f,tc}$	1.66861
$K_{f,b}$	2.00193
$K_{f,t}$	1.308

Table 2.3: Fatigue notch factors

We now know:

$$\sigma_{m,eff} = \sqrt{\left(K_f^b \sigma_{nom}^b + K_f^c \sigma_{nom}^c\right)^2 + 3 \left(K_f^t \tau_{nom}^t\right)^2} = 88.3 MPa$$

$$\sigma_{a,eff} = \sqrt{\left(K_f^b \sigma_{a,nom}^b + \frac{K_f^{tc} \sigma_{a,nom}^{tc}}{0.85}\right)^2 + 3 \left(K_f^t \tau_{a,nom}^t\right)^2} = 87.7 MPa$$

Finally:

$$\sigma_{a,lim} = \sigma_{D-1,C} - M_\sigma \cdot \sigma_{m_{eff}} = 325.16 MPa$$

$$\text{Safety Factor} = \frac{\sigma_{a,lim}}{\sigma_{a,eff}} = 3.7$$

We can now plot the complete Haigh diagram.

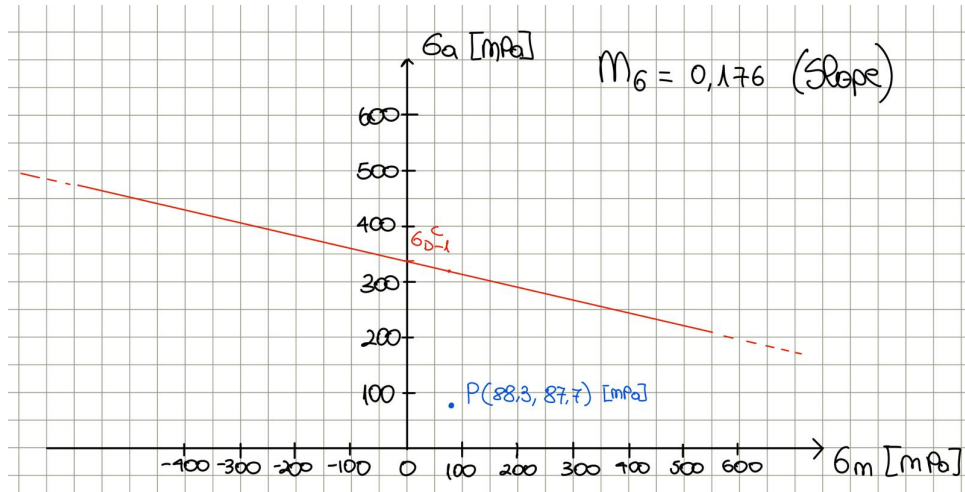


Figure 2.5: Haigh Diagram for section z3

Thanks to this diagram, we see that the working points belongs to the area of the safety factor. So this section can resist infinite life in fatigue.

2.3.2 Section zG - Gear

The same formulas we can use to perform the verification on the gear of the pinion.

Values such as K_a , σ_{adm} , σ_{adp} , K_{dm} , K_{dp} are the same since also the diameter is the same (112.5 mm). Also R_m and R_{eH} are the same. Since the roughness value will change, also $\sigma_{C,D-1}$ will change: By using a value of $b_0=1$, K_s and K_v 1:

$$\sigma_{C,D-1} = 354.9 MPa$$

From the file provided in the class for the shafts with splines and based on the loads, we can use different formulas:

5.3.3.6 Shafts with splines

The fatigue notch factors of a splined or serrated shafts for torsion are to be determined for the nominal stress computed with the innermost diameter either from Figure 5.3.12 or according to the following equation:

$$K_{t,t}(d_p) = \exp \left[4,2 \cdot 10^{-7} \cdot (R_m / \text{MPa})^2 \right] \quad (5.3.19)$$

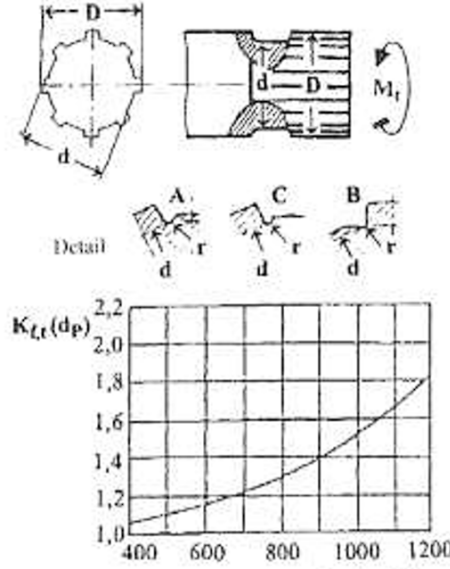


Figure 2.6: Kf for torsion

And we obtain a value of Kf equal to 1.29

Instead, for flexural stress and tension compression case we use the followings:

The fatigue notch factors for bending are to be computed for splined shafts after Eq. (5.3.20) and for serrated shafts after Eq. (5.3.21):

$$K_{t,b}(d_p) = 1 + 0,45 \cdot (K_{t,t}(d_p) - 1), \quad (5.3.20)$$

$$K_{t,b}(d_p) = 1 + 0,65 \cdot (K_{t,t}(d_p) - 1), \quad (5.3.21)$$

For tension-compression approximately the same values apply as for bending.

Figure 2.7: Kf for bending and normal stresses

From these formulas, we obtain for K_f^b and K_f^{tc} equal to 1.194.

We now know:

$$\sigma_{m,eff} = \sqrt{\left(K_f^b \sigma_{nom}^b + K_f^c \sigma_{nom}^c \right)^2 + 3 \left(K_f^t \tau_{nom}^t \right)^2} = 80.3 \text{ MPa}$$

$$\sigma_{a,eff} = \sqrt{\left(K_f^b \sigma_{a,nom}^b + \frac{K_f^{tc} \sigma_{a,nom}^{tc}}{0.85} \right)^2 + 3 \left(K_f^t \tau_{a,nom}^t \right)^2} = 135.6 \text{ MPa}$$

Finally:

$$\sigma_{a,lim} = \sigma_{D-1,C} - M_\sigma \cdot \sigma_{m,eff} = 340.75 \text{ MPa}$$

$$\text{Safety Factor} = \frac{\sigma_{a,lim}}{\sigma_{a,eff}} = 2.51$$

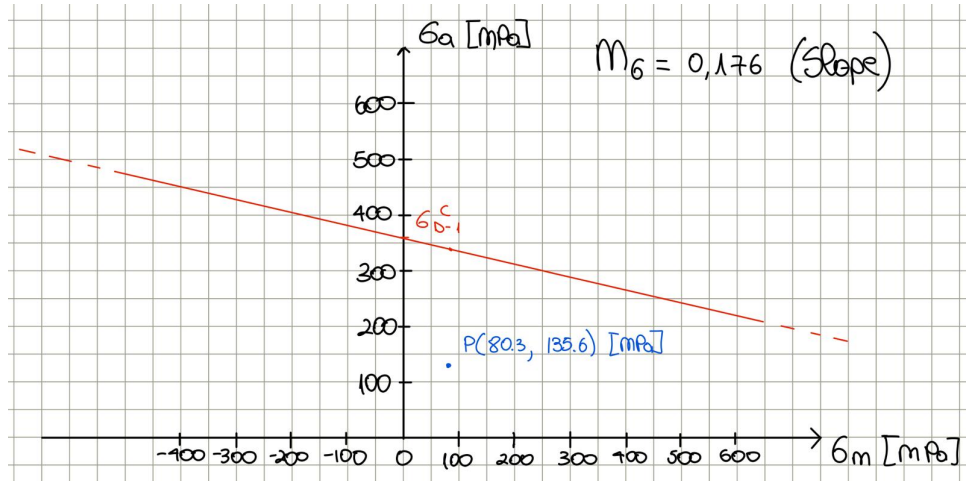


Figure 2.8: Haigh diagram section zG

Therefore, we can say that the section of the gear is more critical for the part of the fatigue, but still in a safe zone

2.4 Shaft 2

The same analogy to verify the material is used for shaft 2.

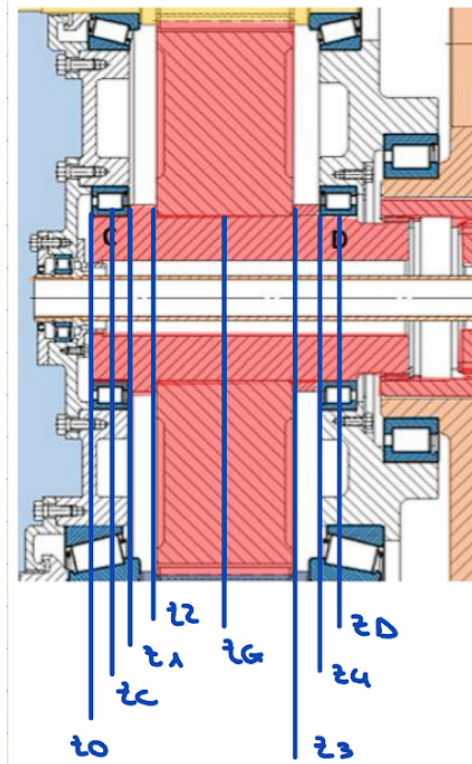


Figure 2.9: Sections of Shaft 2

2.4.1 Section zG - Gear

With the same procedure we used in sections before:

$$a_{d,m} = 0.3d_{eff} = 269.63d_{eff,N} = 16 \quad (3.22)$$

Ka =0.8, ad,m=0.3 We can compute the coefficient $K_{d,m}$.

$$K_{d,m} = \frac{1 - 0.7686 \cdot a_{d,m} \cdot \log(\frac{d_{eff}}{7.5})}{1 - 0.7686 \cdot a_{d,m} \cdot \log(\frac{d_{eff,N}}{7.5})} = \frac{1 - 0.7686 \cdot 0.3 \cdot \log(\frac{269.63}{7.5})}{1 - 0.7686 \cdot 0.3 \cdot \log(\frac{16}{7.5})} \simeq 0.694 \quad (3.23)$$

This coefficient $K_{d,m}$ gives us the ability to evaluate R_m :

$$R_m = 1250 \cdot K_{d,m} = 1250 \cdot 0.798 = 694 \text{ MPa} \quad (3.24)$$

Therefore we now obtain:

$$\sigma_{D-1} = R_m \cdot f_w = 997.76 \cdot 0.45 = 312.3 \text{ MPa} \quad (3.25)$$

With $b_0 = 1$ due to 0.8 micrometer roughness s, we evaluate then $\sigma_{D-1,C}$ and σ_A

$$\sigma_{D-1,C} = \sigma_{D-1} \cdot K_V \cdot K_S \cdot b_0 = 312.2 \text{ MPa} \quad (3.26)$$

$$\sigma_{m,eff} = \sqrt{(\sigma_{nom}^b + \sigma_{nom}^c)^2 + 3(\tau_{nom}^t)^2} = 40.24 \text{ MPa}$$

$$\sigma_{a,eff} = \sqrt{(\sigma_{a,nom}^b + \sigma_{a,nom}^{tc})^2 + 3(\tau_{a,nom}^t)^2} = 13.6 \text{ MPa}$$

We can read from the tables of FKM the values for steel of $a_M = 0.35$ and $b_M = -0.1$, so that we can evaluate:

$$M_\sigma = a_M \cdot 10^{-3} \cdot R_m + b_M = 0.1429$$

he same parameters of materials are applied in shaft 2:

$$a_G = 0.5 \quad b_G = 2700 \quad (3.27)$$

Finally:

$$\sigma_{a,lim} = \sigma_{D-1,C} - M_\sigma \cdot \sigma_{m,eff} = 306.5 \text{ MPa}$$

$$\text{Safety Factor} = \frac{\sigma_{a,lim}}{\sigma_{a,eff}} = 22.6$$

The value shows that our system is very safe

2.4.2 Section Z3- Shoulder

Using always the same procedure, but taking into consideration that here we have the notch effect

$$a_{d,m} = 0.3 d_{eff} = 269.63 \text{ smallest diameter } d_{eff,N} = 16 \quad (3.22)$$

Ka =0.8, ad,m=0.3 We can compute the coefficient $K_{d,m}$.

$$K_{d,m} = \frac{1 - 0.7686 \cdot a_{d,m} \cdot \log(\frac{d_{eff}}{7.5})}{1 - 0.7686 \cdot a_{d,m} \cdot \log(\frac{d_{eff,N}}{7.5})} = \frac{1 - 0.7686 \cdot 0.3 \cdot \log(\frac{269.63}{7.5})}{1 - 0.7686 \cdot 0.3 \cdot \log(\frac{16}{7.5})} \simeq 0.694 \quad (3.23)$$

This coefficient $K_{d,m}$ gives us the ability to evaluate R_m :

$$R_m = 1250 \cdot K_{d,m} = 1250 \cdot 0.798 = 694 \text{ MPa} \quad (3.24)$$

Therefore we now obtain:

$$\sigma_{D-1} = R_m \cdot f_w = 997.76 \cdot 0.45 = 312.3 \text{ MPa} \quad (3.25)$$

With $b_0 = 1$ due to 0.8 micrometer roughness s, we evaluate then $\sigma_{D-1,C}$ and σ_A

$$\sigma_{D-1,C} = \sigma_{D-1} \cdot K_V \cdot K_S \cdot b_0 = 312.2 \text{ MPa} \quad (3.26)$$

Using ratios of r/d 0.0093 and D/d 1.125, we found the values of Kt:

$K_{t,tc}$	2.62
$K_{t,b}$	2.87
$K_{t,t}$	1.76

Table 2.4: Stress concentration factors

To find K_f , we still need the values of n_σ for r and d :

$\chi_{\sigma b}(r)$	0.92	$n_\sigma(r)$	1.167834
$\chi_{\sigma b}(d)$	0.007418	$n_\sigma(d)$	1.004104
$\chi_{\sigma tc}(r)$	0.92	$n_{\sigma tc}(r)$	1.167834
$\chi_{\sigma tc}(d)$	0	$n_{\sigma tc}(d)$	1
$\chi_\tau(r)$	0.46	$\pi_\tau(r)$	1.118677
$\chi_\tau(d)$	0.007418	$\pi_\tau(d)$	1.004104

Now that we have all the values, we can calculate K_f as:

$$K_f = \frac{K_t}{n_{\sigma,r} \cdot n_{\sigma,d}}$$

$K_{t,tc}$	2.24
$K_{t,b}$	2.45
$K_{t,t}$	1.56

$$\sigma_{m,eff} = \sqrt{\left(K_f^b \sigma_{nom}^b + K_f^c \sigma_{nom}^c\right)^2 + 3 \left(K_f^t \tau_{nom}^t\right)^2} = 63 MPa$$

$$\sigma_{a,eff} = \sqrt{\left(K_f^b \sigma_{a,nom}^b + \frac{K_f^{tc} \sigma_{a,nom}^{tc}}{0.85}\right)^2 + 3 \left(K_f^t \tau_{a,nom}^t\right)^2} = 12.8 MPa$$

We can read from the tables of FKM the values for steel of $a_M = 0.35$ and $b_M = -0.1$, so that we can evaluate:

$$M_\sigma = a_M \cdot 10^{-3} \cdot R_m + b_M = 0.1429$$

he same parameters of materials are applied in shaft 2:

$$a_G = 0.5 \quad b_G = 2700 \quad (3.27)$$

Finally:

$$\sigma_{a,lim} = \sigma_{D-1,C} - M_\sigma \cdot \sigma_{m,eff} = 303.3 MPa$$

$$\text{Safety Factor} = \frac{\sigma_{a,lim}}{\sigma_{a,eff}} = 23.7$$

The value shows that our system is very safe

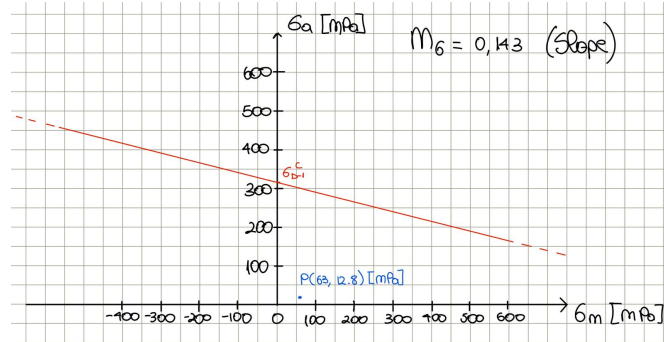


Figure 2.10: Haigh diagram shoulder on the shaft 2

Chapter 3

Part 3 - Bearings

The following chapter is dedicated to the evaluation of the dynamic bearing load P for the three following bearings:

Verification of Bearing Rating Life

This document summarizes the data required to verify the rating life of the bearings C, D, and E under two different loading conditions. The necessary bearing data can be retrieved from the SKF online catalogue, available at:

<https://www.skf.com/group>

Bearings and Bore Diameter

Bearing	Bore Diameter (d)
Bearings C and D - series NCF 2952 CV	260 mm
Bearing E - series 22338 CC/W33	190 mm

Additional Data

Consider also the following operating conditions and parameters:

- **Lubrication type:** circulating oil
- **Lubricant:** Synthetic oil with viscosity ISO VG 320
- **Contamination factor:** $\eta_c = 0.5$ (suggested)
- **Operating temperature:** $T = 50$ °C
- **Minimum requested bearing life:** $L_h = 50\,000$ operating hours (requested reliability 96%)
- **Total transmission ratio:** 41.63

For what concerns the evaluation of a_{SKF} through the viscosity ratio k , we will seek values of k such that: $1 < k < 4$. However, this condition will not always be respected, especially in the most excited bearings.

We have to verify the life of the bearings under two working conditions.

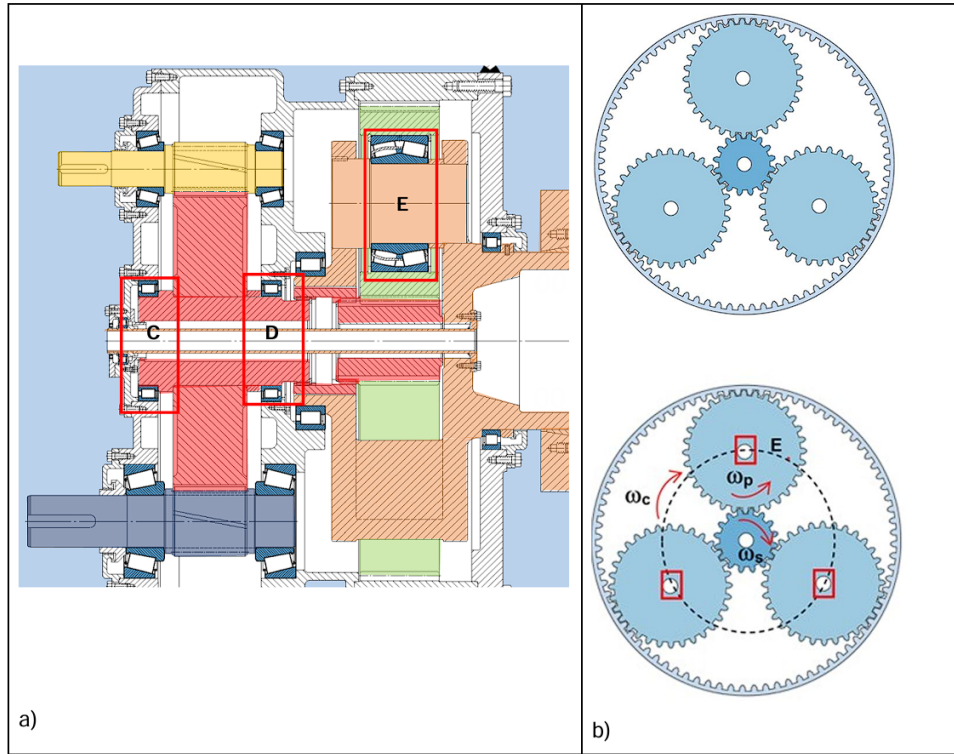


Figure 3.1: Power and speed (at input) along one year

3.1 Loading condition 1

We have to calculate the rating life (in hours) of the bearings: C, D and E assuming full power working conditions (already used in computations against static and fatigue failure of the speed increaser shafts in Part 1 and Part 2):

- **Input power** (P_t): 1150 kW
- **Input speed** (n_t): 24.5 rpm

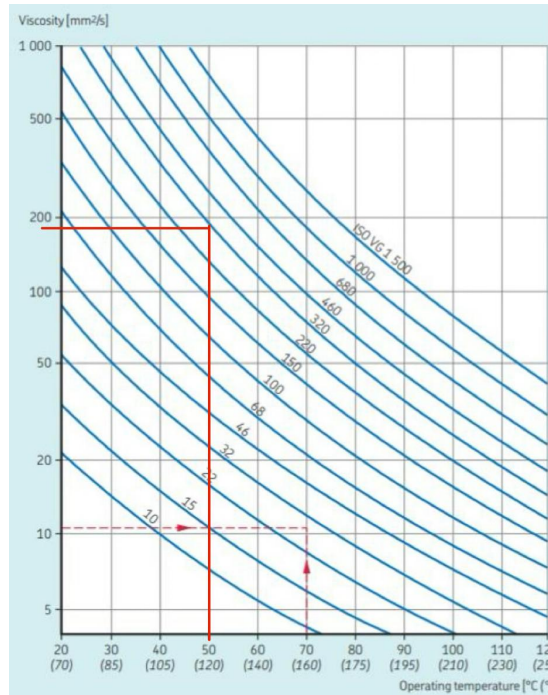
Bearing C

- first thing first it's needed the evaluation of forces supported by the bearing, in this case it can only support a radial reaction which is obtained as the vectorial summation of the two components already obtained in the static analysis as reaction of pin C.

$$F_{r,C} = \sqrt{R_{C,y}^2 + R_{C,x}^2} = 97224 \text{ N} \quad (4.1)$$

Now it's possible to evaluate the equivalent dynamic load P, in this case it's simply $P = 0.4F_{r,C} = 38889.6 \text{ N}$ since the bearing can not support an axial component.

- the second step is the evaluation of 'viscosity ratio' $k = \frac{v}{v_1}$ using the following diagrams. In order to do so we need to remind the values of bore diameter $d = 260 \text{ mm}$ and outer diameter $D = 360 \text{ mm}$. The coefficient v (Operating kinematic viscosity) has been picked knowing the max operating temperature $T = 50^\circ\text{C}$ and the lubricant ISO VG 320.



The value results to be $v = 190 \text{ mm}^2/\text{s}$, and it does not depend on the type of bearing (equal for C and D).

The coefficient v_1 (Rated kinematic viscosity) is found using the mean diameter $d_m = \frac{d+D}{2} = 310\text{ mm}$ along with the angular speed of shaft 2: $n_2 = \frac{n_1}{i_1} = 132.7\text{ rpm}$ (i_1 is the transmission ratio).

It can be read that $v_1 = 48 \text{ mm}^2/\text{s}$. And so $k = \frac{v}{v_1} = 3.96$, which will be used to read the value of a_{SKF} later on.

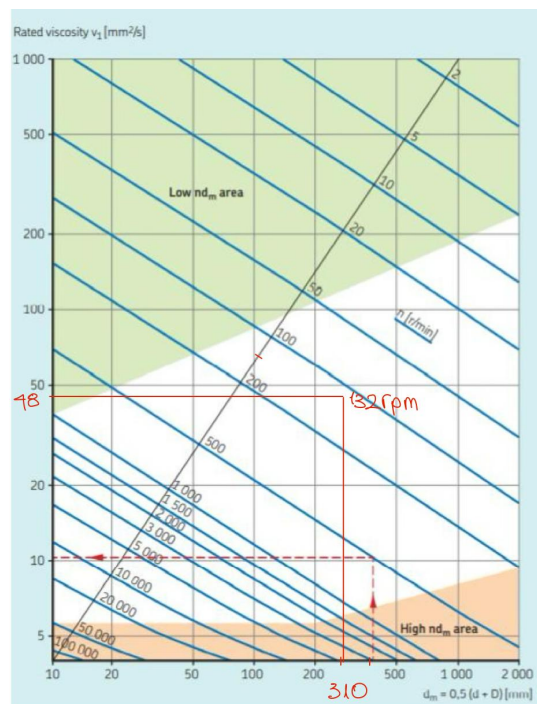


Figure 3.3: Rated kinematic viscosity

- Now we can read on the SKF Handbook for the bearing C (Series NCF 2952 CV) the values of:

Fatigue load limit	P_u	143 kN
Basic dynamic load rating	C	737 kN

also we need to set $p = \frac{10}{3}$ since bearing C is a roller bearing, finally remember that $\eta_c = 0.2$ (contamination factor).

- Now we can evaluate the rating life in millions of revolution using

$$L_{nm} = a_1 a_{SKF} \left(\frac{C}{P} \right)^p \quad (4.2)$$

where p, C, P are already known, $a_1 = 1$ using a **reliability** of 90%.

Considering that bearing C belongs to Series NCF 2952 CV and from SKF Handbook is said that it's also an **EXPLORER** bearing, we evaluate $\eta_c \cdot \frac{P_u}{P} = 1.84$, $k = 3.95$ and we can read from the diagram that a_{SKF} has a very high value.

Values for life adjustment factor a_1			
Reliability	Failure probability	SKF rating life	Factor
%	%	L_{nm} million revolutions	a_1
90	10	L_{10m}	1
95	5	L_{5m}	0,64
96	4	L_{4m}	0,55
97	3	L_{3m}	0,47
98	2	L_{2m}	0,37
99	1	L_{1m}	0,25

Figure 3.4: Life adjustment factor

With all this information we conclude that

$$L_{10m} = a_1 a_{SKF} \left(\frac{C}{P} \right)^p = 838.3 \quad \text{million of cycles} \quad (4.3)$$

$$L_{10m,h} = \frac{L_{nm} \cdot 10^6}{60 \cdot n_2} = 105303.2 \quad \text{hours} \quad (4.4)$$

where $n_2 = 580 \text{ rpm}$ is the angular speed of shaft 2.

Since the requested minimum life $L_{10mh} = 50000 \text{ hours}$ we can conclude that bearing C is widely **VERIFIED**.

3.2 Bearing D

Bearing D is same as C, but it has to sustain also axial load which has the direction towards bearing D, therefore bearing D has to sustain radial and axial load.

- first thing first it's needed the evaluation of forces supported by the bearing, in this case we have both radial and axial reaction. The radial reaction is obtained as the vectorial summation of the two components already obtained in the static analysis for the bearing D;

$$F_{r,D} = \sqrt{R_{D,y}^2 + R_{D,z}^2} = 133888.55 \text{ N} \quad (4.5)$$

while the axial reaction is already known without calculations.

$$F_{a,D} = R_{D,z} = 50458.21 \text{ N} \quad (4.6)$$

Before going on we need to take from the SKF Handbook the following coefficient for our bearing:

Limiting value, e	0.3
Axial load factor, Y_a	0.4
Radial load factor, Y_r	0.92

To evaluate the equivalent dynamic load P we need to evaluate

$$\frac{F_{a,D}}{F_{r,D}} = 0.38 > 0.35 \quad (4.7)$$

Then we can see that this ratio is higher than $e = 0.3$, so to compute the equivalent dynamic load P , in this case we use:

$$P = Y_r \cdot F_{r,D} + Y_a \cdot F_{a,D} = 143360.75 \text{ N} \quad (4.8)$$

- the second step is the evaluation of 'viscosity ratio' $k = \frac{v}{v_1}$ using the following diagrams. In order to do so we need to remind the values of bore diameter $d = 260 \text{ mm}$ and outer diameter $D = 360 \text{ mm}$. The coefficient v (Operating kinematic viscosity) has been picked knowing the max operating temperature $T = 50^\circ\text{C}$ and the lubricant ISO VG 320.

The coefficient v_1 (Rated kinematic viscosity) is found using the mean diameter $d_m = \frac{d+D}{2} = 310 \text{ mm}$ along with the angular speed of shaft 2: $n_2 = \frac{n_1}{i_1} = 132.7 \text{ rpm}$ (i_1 is the transmission ratio).

It can be read that $v_1 = 48 \text{ mm}^2/\text{s}$. And so $k = \frac{v}{v_1} = 3.96$, which will be used to read the value of a_{SKF} later on.

- Now we can read on the SKF Handbook for the bearing D (Series 22326CC/W33) the values of:

Fatigue load limit	P_u	143 kN
Basic dynamic load rating	C	737 kN

also we need to set $p = \frac{10}{3}$ since bearing D is a roller bearing, finally remember that $\eta_c = 0.2$ (contamination factor).

- Now we can evaluate the rating life in millions of revolution using

$$L_{nm} = a_1 a_{SKF} \left(\frac{C}{P} \right)^p \quad (4.9)$$

where p, C, P are already known, $a_1 = 1$ using a **reliability** of 90%.

Considering that bearing D is same as C series and from SKF Handbook is said that it's also an **EXPLORER** bearing, we evaluate $\eta_c \cdot \frac{P_u}{P} = 0.499$, $k = 3.95$ and we can read from the diagram that $a_{SKF} = 6.5$.

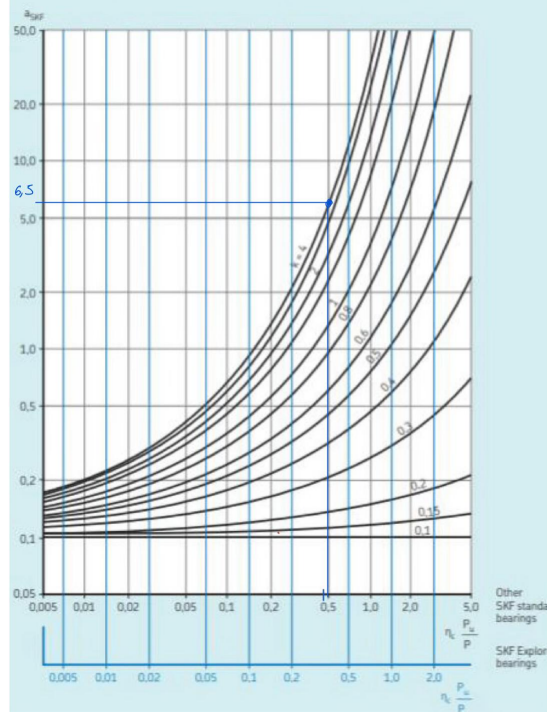


Figure 3.5: Factor a_{SKF}

With all this information we conclude that

$$L_{10m} = a_1 a_{SKF} \left(\frac{C}{P} \right)^p = 838.3 \quad \text{million of cycles} \quad (4.10)$$

$$L_{10,h} = \frac{L_{nm} \cdot 10^6}{60 \cdot n_2} = 105303.2 \quad \text{hours} \quad (4.11)$$

where $n_2 = 132.7 \text{ rpm}$ is the angular speed of shaft 2.

Since the requested minimum life $L_{10mh} = 50000 \text{ hours}$ we can conclude that bearing D is also widely **VERIFIED**.

3.3 Bearing E

Bearing E is used to support the planet of the planetary gearbox. The sketch shows that actually the shaft is supported by tapered roller bearings of series 22338 CC/W33 . It is our goal to derive, before we analyze the P and fatigue, the radial force applied on each bearing and the some velocities.

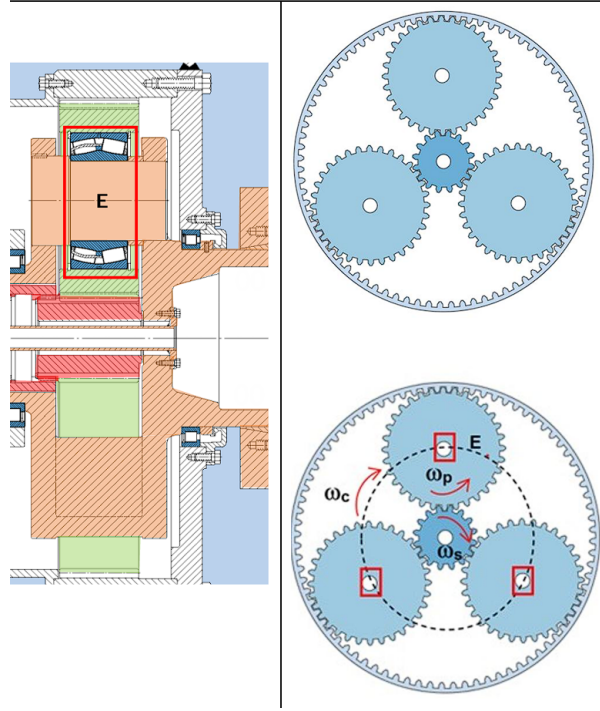


Figure 3.6: Planet of the planetary gearbox

It is important to highlight that the planetary gearbox, since involves only spur gears, does not produce any axial load. Therefore, only a radial load will be applied to the whole shaft.

3.3.1 Dynamic Load

In order to derive properly all the quantities we need, we try to comment a schematic design of the planetary gearbox, deriving mainly the actual force applied to each bearing and the velocity of rotation of the planet, which is equal to the velocity of rotation of bearing E.

To compute the force, we will refer to figure 4.11:

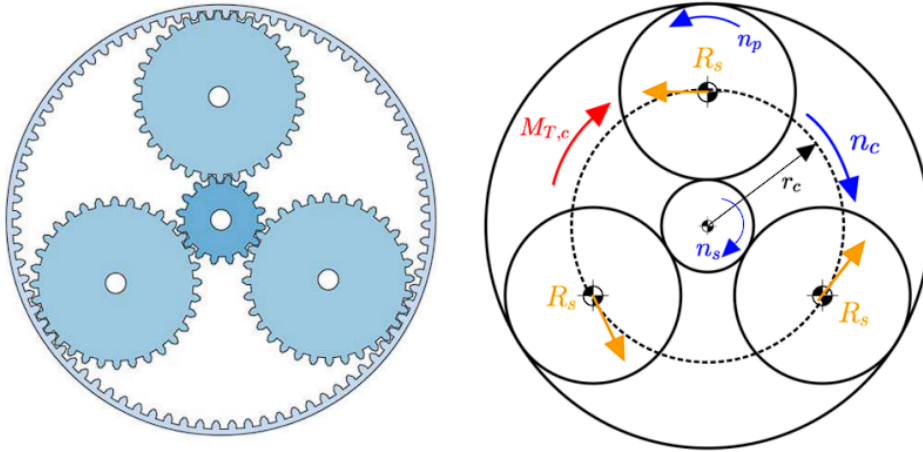


Figure 3.7: Planetary gearbox: schematic sketch

The depicted circles at the right are referring to the **primitive diameters**. Since the configuration is completely symmetric, the reactions of the shafts R_s . We are referring with R_s to the force applied to the whole shaft, which is different from the force applied to the single bearing. By denoting with $M_{T,c}$ the torque applied at the **carrier**, and by writing an equilibrium of moment upon the carrier itself, we

derive:

$$R_s = \frac{M_{T,c}}{3r_c} \quad (4.12)$$

Where r_c is the primitive radius of the carrier. Finally, by considering that $M_{T,c}$ is the torque applied at the carrier, which is the output shaft, we derive:

$$M_{T,c} = \frac{P_{in} \cdot 60}{2 \cdot \pi \cdot n_{in}} = 448232.3 Nm$$

P_{in} refers to the input power. By considering that the radius of the carrier is obtained by:

$$r_c = r_s + r_p$$

Where r_p is the radius of the planet, whereas r_s is the radius of the sun. The following definitive equation can be obtained:

$M_{T,c} [Nm]$	$P_{in} [kW]$	$r_s [mm]$	$r_p [mm]$	$R_s [N]$
448232.3	1150	94.3	232.5	457118.9

For the given type of bearing, the SKF handbook suggest the following value for the **dynamic bearing load** P :

$$P = R_s = 457118.9 [N]$$

3.3.2 Velocities in the planetary gearbox

Before we analyze the fatigue of bearing E, we must evaluate what is the spinning velocity of the bearing. By figure down, it can be stated that the spinning speed of the bearing coincides with the spinning speed of the planet. Therefore, we must compute the speed of the planetary. From the main sketch it is possible to see that the outer **annulus** is fixed with the casing, therefore it is fixed ($v = 0$). The sun instead coincides with shaft 2 (the red shaft), the carrier with the output shaft (orange) and the planet with the green gears. By depicting again the contact zone at the primitive diameters, and now we can derive a sketch where we highlight the velocities we are interested in:

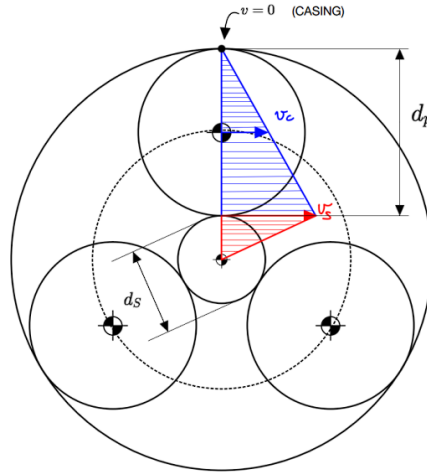


Figure 3.8: Velocities of the planetary

We can derive the angular velocity of the planet by the following analogy:

$$v_p = v_s$$

$$\omega_p \cdot r_p = \omega_s \cdot r_s$$

$$\text{Instead : } v_c = \omega_c \cdot r_c = \omega_c \cdot (r_s + r_p)$$

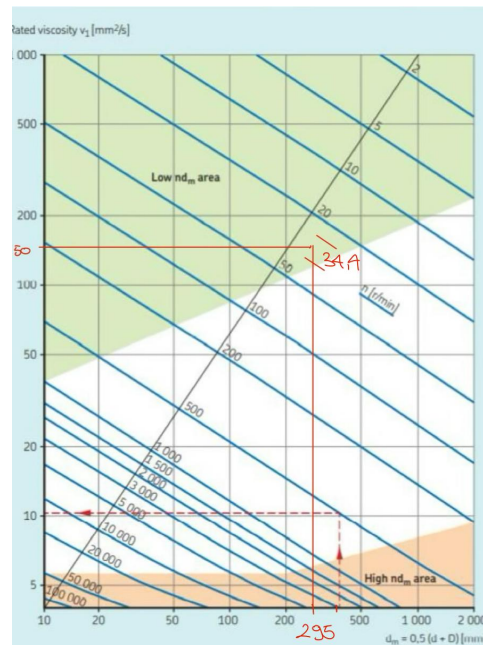
So we can get the speed of the planet:

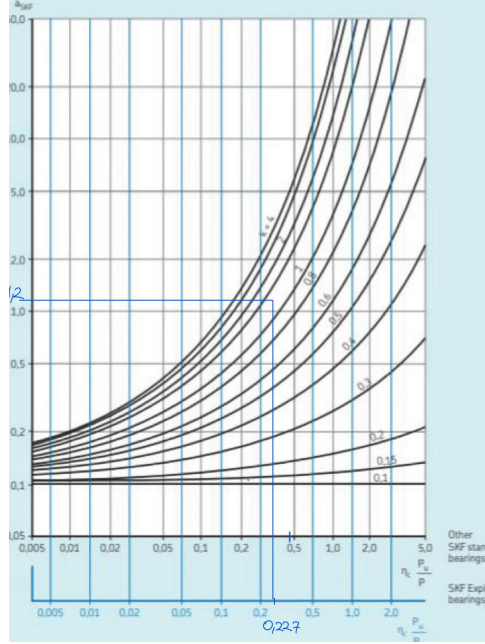
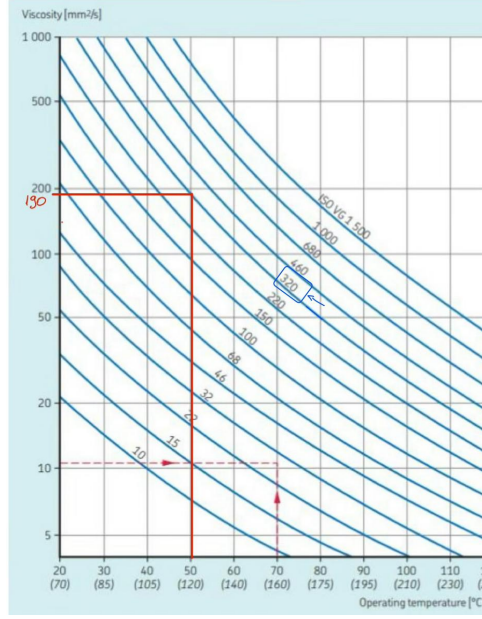
$$\omega_p = \frac{\omega_C \cdot (r_s + r_p)}{r_p} = 34.44 \text{rpm}$$

Parameter	Value	Unit
d	190	mm
D	400	mm
d_m	295	mm
ν_1	150	mm^2/s
ν	190	mm^2/s
k	1.266667	–
C	2120	kN
C_0	2400	kN
p	3.333333	–
a_{SKF}	1.2	–
P_u	208	kN
$\eta_c \times \left(\frac{P_u}{P}\right)$	0.227512	–

Table 3.1: Bearing and lubrication parameters

The values were obtained by the following graphs:





In this case, bearing E, we have tapered roller bearings, therefore p is $10/3$. Since the requested **reliability** is of 90%, $a_1 = 1$. The dynamic load P was previously computed, and it is 457 [kN], whereas the **basic dynamic loading rating**, or dynamic loading capacity, is given by the catalogue, also depicted in figure 4.10, and is $C = 280$ [kN].

The same procedure and graph were used to obtain this table. With all this information we conclude that

$$L_{10m} = a_1 a_{SKF} \left(\frac{C}{P} \right)^p = 109.8 \quad \text{million of cycles} \quad (4.10)$$

$$L_{10m,h} = \frac{L_{nm} \cdot 10^6}{60 \cdot n_2} = 74687.6 \quad \text{hours} \quad (4.11)$$

where $n_2 = 132.7 \text{ rpm}$ is the angular speed of shaft 2.

Since the requested minimum life $L_{10mh} = 50000 \text{ hours}$ we can conclude that bearing D is also widely **VERIFIED**.

3.3.3 Variable working condition analysis-Loading Condition 2

Now that we verified that all bearings are capable of supporting the required 50'000 hours, we can also see that bearing E is the most critical one, with a duration that is very close to the minimum value of 50'000, at least if compared with the other two bearings.

Therefore, if we want to analyze the variable working condition of the bearing, we can simply analyze bearing E, since it is the most critical bearing.

So we are focusing on the bearing with the shortest rating life (in hours) and, **for this bearing only**, we can consider variable working conditions by keeping into account that the wind turbine is a variable speed turbine.

Refer to the graph of Figure 3.9 to compute the torque and the speed at the different shafts in each working condition.

Assume that the generator power fraction and speed fraction that you read on the graph also apply to the variation in input power and input speed from the nominal working condition W1.

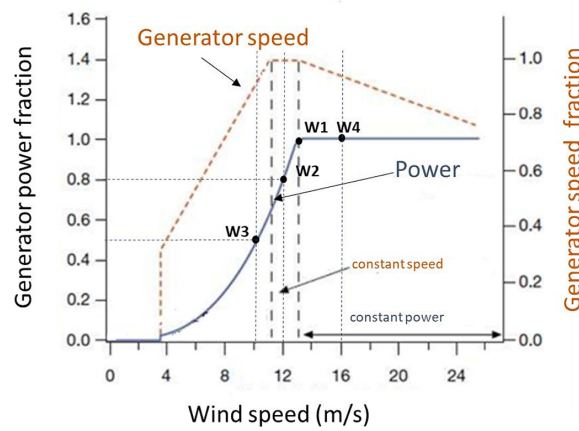


Figure 3.9: Typical generator power and speed fraction

Working Condition	Time [%]
W1	60
W2	30
W3	8
W4	2

Table 3.2: Working conditions and time fractions

- We have to compute the SKF rating life (in hours) of the bearing under these variable loading conditions, using Miner's rule:

$$L_{10m} = \frac{1}{\sum_i \frac{U_{norm,i}}{L_{10mi}}}$$

where:

- U_i are the life fractions (in cycles) under the different load conditions (we have to transform the given time percentages into percentages in cycles).
- L_{10mi} are the SKF rating lives under the different load conditions.

- We have to calculate the obtained SKF rating life in hours and verify whether it is higher than the required minimum bearing life $L_{10h} = 50\,000$ operating hours.

L_{10mi} is the duration of the bearing (expressed as number of cycles) as if it was under the only loading condition i , and U_i is the **fraction of life**, which is different from the fraction of time. U_i can be derive by the following formula:

$$U_i = \%T_i \cdot n_i$$

Table 3.3: Performance under Nominal Conditions

Parameter	Nominal	w1	w2	w3	w4
Power (kW)	1150	1150	920	575	1150
Input Speed (rpm)	24.5	24.5	24.5	22.3	22.54
Torque (N m)	487 209	448 232.3	358 585.8	246 281.5	487 209
Force (N)	496 868.4	457 118.9	365 695.1	251 164.3	496 868.4
Dynamic Bearing Load (N)	496 868.4	457 118.9	365 695.1	251 164.3	496 868.4

Where Torques are calculated as:

$$M_{T,i} = \frac{P_{in} \cdot 60}{2 \cdot \pi \cdot n_{in}}$$

And Forces are calculated as:

$$F = \frac{M_{T,i}}{3r_c} \quad (4.12)$$

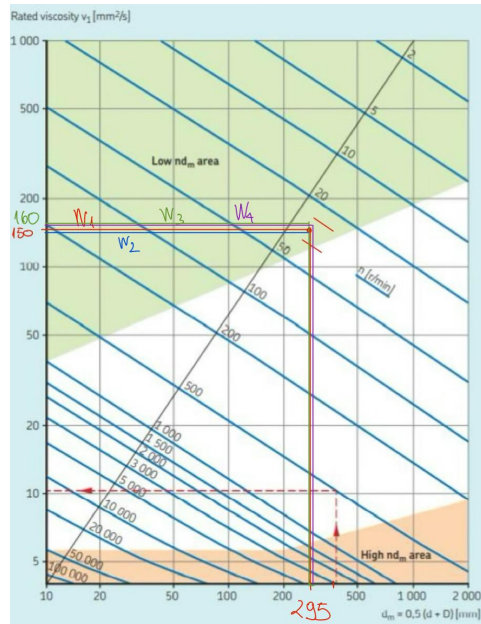
Now, we have to compute the different L_{10mi} . By considering a_1 , by recalling back the equation we have to evaluate, for each working condition, the a_{SKF} factor. From below, we can see that:

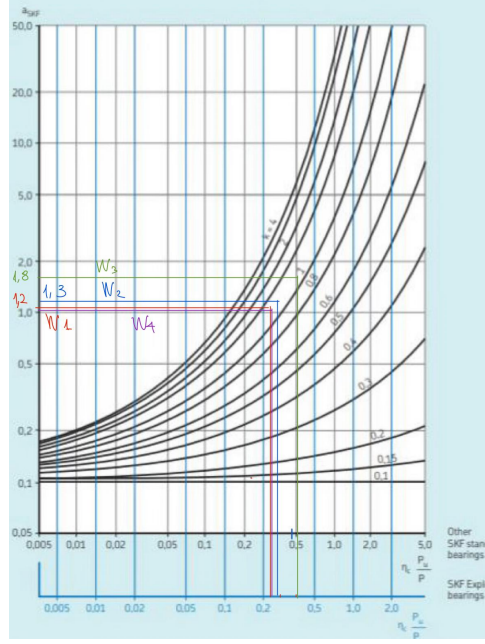
$$a_{SKF} = f(P, k)$$

With $k = f(n)$. So by changing the load P and the spin speed n , the coefficients are changing as well.

$$L_{10mi} = a_1 a_{SKF} \left(\frac{C}{P} \right)^p$$

$$L_{10mi,h} = \frac{L_{nm} \cdot 10^6}{60 \cdot n_i}$$





Parameter	W1	W2	W3	W4
Input Speed [rpm]	24.5	24.5	22.295	22.54
Planet Speed [rpm]	34.44	34.44	31.3404	31.6848
v_1 [mm ² /s]	150	150	160	157
k	1.27	1.27	1.1875	1.21
$\eta_c \times (P_u/P)$	0.23	0.28	0.41	0.23
a_{SKF}	1.2	1.3	1.8	1.2
L_{10mi} [10 ⁶ revs]	109.8	250.2	1212.2	83.1
L_{10mi} [hours]	74 687.6	170 233.2	906 156.6	74 687.6
U_i	14.7	7.35	1.8	0.45
U_{tot}	24.3			
U_{norm}	0.605 327	0.302 663 438	0.073 446	0.018 563
L_{10m} [hours, total]	97 921.01			

Table 3.4: Computed parameters under working conditions W1–W4

The **bearing E** is **VERIFIED** for these working conditions. We obtain a number higher than 50'000 hours.

Chapter 4

Part 4 - Bearings Assembly

In this brief chapter we will consider the bearing arrangements of the input shaft (shaft 1), and we will carry out a detailed analysis of the bearing arrangements. Namely we are going to discuss:

- which bearing is locating and which is non-locating,
- their mounting type,
- and finally we will list and explain the various functions of the relevant parts for the functioning and mounting of the whole arrangement.

4.1 Bearing analysis

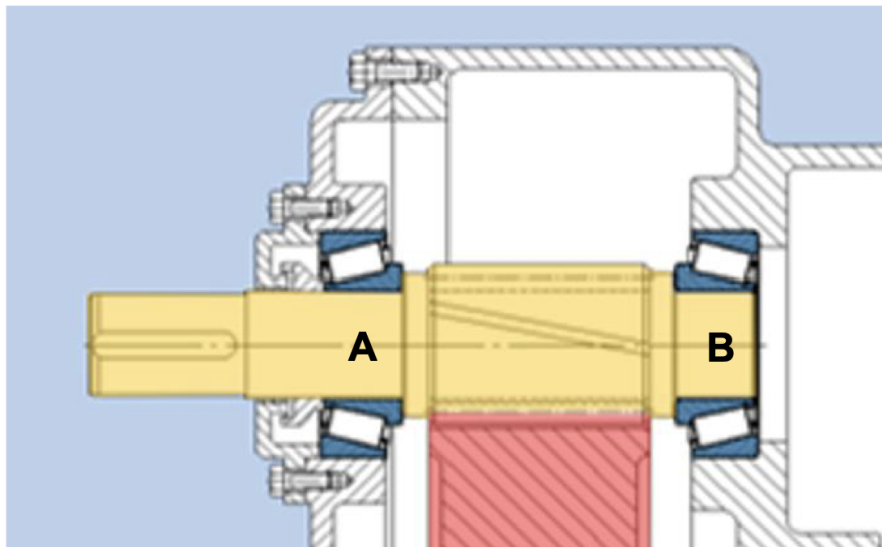


Figure 4.1: shaft 1

With reference the figure 5.1 we can observe that **bearing A is the locating** one since the direction of the axial force makes it provide an axial reaction. In reality, since the layout of the system allows to change the angle of the Blades, changing the verse of rotation, the configuration is a cross locating one, since the back to back arrangement allows us to have a different locating condition, depending on the sense of rotation, and so the verse of axial force. Since we initially considered a verse of rotation so that the axial thrust was going to left, the bearing A is now locating. For both the two bearings the inner ring is mounted with interference fit because they are connected to the rotating shaft; the outer ring of bearing A is mounted with slide fit since it's the place were there could be pre-load registration if it's needed.

Bearing B is the non-locating one because it has the aim of accommodating the axial displacement due to thermal expansion of the shaft and compensate for the accumulation of tolerances of components (which could affect the distance between the two bearings).

So, the **outer ring** of bearing B is mounted with **slide fit** because the non-locating bearing must provide axial displacement of the shaft.

Here below there is a summary of what we already explained.

ID	Component type	Description
A	single row tapered roller bearings	Inner ring: The interference fit allows it to rotate coherently with the shaft. Outer ring: The slide fit allows the preload application and the adaptation of the bearing to the thermal expansion.
B	single row tapered roller bearings	Inner ring: The interference fit allows it to rotate coherently with the shaft. Outer ring: The slide fit allows the preload application and the adaptation of the bearing to the thermal expansion.

Table 4.1: Description of bearings A and B

4.2 Configuration details

Herein are showed the most relevant parts of the whole assembly with a short explanation of their function inside the mounting of shaft 1.

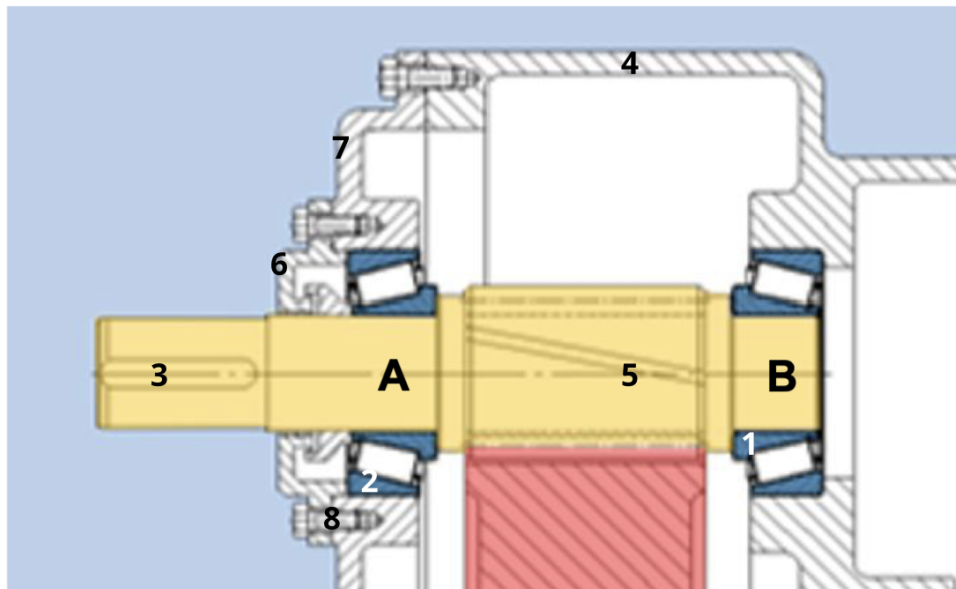


Figure 4.2: Part indexes of shaft 1

Component number	Component type	Function
1	single row tapered roller bearings	non locating bearing, due to cross locating arrangement, allows the axial degree of freedom and the availability of axial displacement of the shaft
2	single row tapered roller bearings	locating bearing, bears the axial thrust due to the force exchange due to the mating of the helical gears
3	Shaft part with keyway	transmits the torque provided to the shaft 1 to the final user, thanks to the key in the keyway
4	casing	protects the system from any external contamination, positioning the system
5	helical gear pinion	allows torque transmission thanks to mating action with the gear belonging to shaft 2
6	flange	apply preload if needed, positioning the outer ring, closes the system
7	cover	closes and covers the system
8	bolts (x2)	physically allows the connection between 6 and 7

Table 4.2: Bill of materials for shaft assembly

4.2.1 Suggested Alternative Layout

The last task was about suggesting an alternative layout, substituting the previous bearings with some others, with different specifics, but obtaining the same kind of reaction overall. Since the task requested the utilization of the axial thrust bearings, our layout is comprehensive of 2 radial roller bearings, one for each side, and 1 double action axial thrust bearing. The rollers are still used, even in the radial arrangement, since the shaft is exposed to an High radial force, and considering that the previous layout used the rollers, even if they have a lower speed capacity, they represent still a valid solution. Anyway, an axial reaction is now needed: For this reason we decided to put a double action axial thrust bearing on the left end of the layout, since it can provide an axial reaction in both the verses. This condition makes the layout similar, in the point of view of the useful effect, to the previous one, since we can simulate the cross locating condition by the usage of that kind of bearing, providing the right reaction for any condition and any verse of rotation.

Chapter 5

Gear Design and Verification

5.1 Estimation of Characteristic Dimensions

From the data we know the overall transmission ratio i_{tot} , and since in the previous section we calculated the transmission ratio of the planetary gear $i_1 = 6.15$, we can now evaluate the transmission ratio of the second stage:

$$i_2 = \frac{i_{tot}}{i_1} = \frac{41.63}{6.15} = 6.77 \quad (5.1)$$

Then we chose a number of teeth for the pinion (1), making sure it was bigger than the minimum number of teeth (which was found to be of 15), and then we found the number of teeth of the gear (2) thanks to the transmission ratio:

$$z_1 = 19 \quad z_2 = 129$$

Next we chose the normal modulus m_n using the dimensions we measured on the drawing, together with the following equations:

$$p_t = \frac{\pi \cdot d_1}{z_1} = 18.83 \text{ mm}$$

$$p_n = p_t \cdot \cos(\beta) = 18.19 \text{ mm}$$

$$m_n = \frac{p_n}{\pi} = 5.79 \text{ mm}$$

By looking at the table in Figure 5.1 we have selected a normal modulus of 6 mm.

Table 3.2 Standard values of module Unit: mm

I	II	I	II
0.1	0.15	3	3.5
0.2	0.25	4	4.5
0.3	0.35	5	5.5
0.4	0.45	6	(6.5)
0.5	0.55		7
0.6	0.7	8	9
	0.75	10	11
0.8	0.9	12	14
1	1.125	16	18
1.25	1.375	20	22
1.5	1.75	25	28
2	2.25	32	36
2.5	2.75	40	45
		50	

*Extracted from JIS B 1701-2: 1999 Cylindrical Gears for general engineering and for heavy engineering - Part 2 : Modules.

Figure 5.1: Table 3.2

With this it is possible to compute the actual pitch radii and addendum radii:

$$\begin{cases} r_1 = 59 \text{ mm} \\ r_2 = 401 \text{ mm} \\ r_{a1} = 65 \text{ mm} \\ r_{a2} = 406.8 \text{ mm} \end{cases} \quad (5.1)$$

Finally the face width was estimated to be:

$$b = 193 \text{ mm}$$

5.2 Bending Fatigue (Pinion)

First, we calculated the specific pitch-line force using the following equation:

$$w_{Ft} = w \cdot K_I \cdot K_V \cdot K_{F\alpha} \cdot K_{F\beta} \quad (5.2)$$

w is the tangential load k_p on the pitch circle divided by the width:

$$w = \frac{k_p}{b} = 984.5 \text{ N/mm} \quad (5.3)$$

The overload factor K_I was assumed to be 1.25, while the dynamic factor K_V was computed as:

$$K_V = 1 + \left(1 + \frac{k_e \cdot f_{pe}}{w \cdot K_I} 0.65 \text{ N/mm} \right) \cdot C_V \cdot C_\beta \cdot v \quad (5.4)$$

f_{pe} was found using Table A21.6 (Figure 5.2). We considered a manufacturing level number equal to 5, and thus:

$$f_{pe} = 7 \text{ } \mu\text{m}$$

Tab. A21.6 Zulässige Teilungs- und Eingriffsteilungs-Abweichungen für Verzahnungen nach DIN 3962 (Auszug)

d in mm		m_n in mm		f_a und f_{pe} in μm bei Verzahnungsqualität									
über	bis	über	bis	4	5	6	7	8	9	10	11	12	
	10	von 1	2	3	4,5	6	9	12	18	28	45	71	
10	50	von 1	2	3,5	5	7	9	14	18	28	50	80	
		2	3,55	3,5	5	7	10	14	20	32	50	80	
		3,55	6	4	6	8	11	16	22	36	56	90	
		6	10	5	7	10	12	18	25	40	63	110	
50	125	von 1	2	4	5	7	10	14	20	32	50	80	
		2	3,55	3,5	5	7	10	14	20	32	50	80	
		3,55	6	4	6	9	12	16	25	40	63	100	
		6	10	5	7	10	14	20	28	45	71	110	
125	280	von 1	2	4	5,5	8	11	16	22	36	56	90	
		2	3,55	4	6	8	11	16	22	36	56	90	
		3,55	6	4,5	7	9	12	18	25	40	63	100	
		6	10	5,5	8	11	14	20	28	45	71	125	
280	560	von 1	2	4,5	6	8	12	16	22	36	56	100	
		2	3,55	4	6	8	12	16	22	36	56	90	
		3,55	6	5	7	10	14	20	28	45	71	110	
		6	10	6	8	11	16	22	32	50	80	125	
560	1000	von 1	2	4,5	6	8	12	16	22	36	56	90	
		2	3,55	4	6	8	12	16	22	36	56	90	
		3,55	6	5,5	8	11	16	20	28	45	75	125	
		6	10	6	9	11	16	25	32	56	90	140	
1000	1600	von 1	2	4,5	6	8	12	16	22	36	56	90	
		2	3,55	4	6	8	12	16	22	36	56	90	
		3,55	6	5,5	8	11	16	20	28	45	75	125	
		6	10	6	9	11	16	25	32	56	90	140	
1600	2000	von 1	2	4,5	6	8	12	16	22	36	56	90	
		2	3,55	4	6	8	12	16	22	36	56	90	
		3,55	6	5,5	8	11	16	20	28	45	75	125	
		6	10	6	9	11	16	25	32	56	90	140	

Figure 5.2: Table A21.6

k_e was found in table 21.10 (Figure 5.3), and it depends on the teeth material, which in this case is steel on steel, thus:

$$k_e = 4 \mu m^{-1}$$

Tab.21.10 Eingriffsfaktor f_E von Zahnradpaaren

Eingriffsfaktor $f_E \approx k_e \cdot f_{pe} - 8$ f_{pe} = zulässige Eingriffsteilungs-Abweichung für das Rad 2 nach Tab.A21.6					
k_e in μm^{-1} bei Werkstoffpaarung					
St/St	St/GG	GG/GG	St/GGG	GG/GGG	St/Hgw
4	2,3	1,8	3,2	1,3	0,2

Figure 5.3: Table 21.10

From table 21.9 in figure 5.4, we can find C_V (using a pitch velocity of $v_p = 6.45 \text{ m/s}$):

$$C_V = 0.023 \text{ s/m}$$

Tab.21.9 Anhaltswerte für den Dynamikfaktor K_V von Zahnradpaaren

Dynamikfaktor k_e in μm^{-1} f_{pe} in μm w in N/mm K_I C_V in s/m C_β v in m/s		$K_V \approx 1 + \left(1 + \frac{k_e \cdot f_{pe}}{w \cdot K_I} 0,65 \text{ N/mm}\right) C_V \cdot C_\beta \cdot v$ Paarungsbeiwert nach Tab.21.10, Seite 470, zulässige Eingriffsteilungs-Abweichung für das Rad 2 nach Tab.A21.6, spezifische Nennumfangskraft = F_t/b , Betriebsfaktor nach Tab.21.3 Seite 445, Belastungsbeiwert (untenstehend), Schrägungsbeiwert (untenstehend), abhängig von der Sprungüberdeckung ε_β (Gl.21.41), Umfangsgeschwindigkeit der Teilkreise.										
$w \cdot K_I$ N/mm		1	2	4	6	8	10	12	14	16	18	20
25	0,057	0,056	0,056	0,046	0,034	0,029	0,026	0,024	0,021	0,019	0,017	0,017
50	0,050	0,049	0,048	0,041	0,034	0,028	0,025	0,023	0,021	0,019	0,017	0,017
100	0,038	0,037	0,032	0,030	0,027	0,025	0,024	0,023	0,021	0,019	0,017	0,017
200	0,033	0,030	0,025	0,024	0,023	0,023	0,023	0,022	0,021	0,019	0,017	0,017
300	0,031	0,029	0,025	0,023	0,023	0,023	0,023	0,022	0,021	0,019	0,017	0,017
400	0,030	0,029	0,025	0,023	0,023	0,023	0,023	0,022	0,021	0,019	0,017	0,017
500	0,030	0,029	0,024	0,023	0,023	0,023	0,023	0,022	0,021	0,019	0,017	0,017
1000	0,029	0,028	0,024	0,023	0,023	0,023	0,023	0,022	0,021	0,019	0,017	0,017
2000	0,028	0,028	0,024	0,023	0,023	0,023	0,023	0,022	0,021	0,019	0,017	0,017
ε_β		$\leq 0,4$	0,5	0,6	0,7	0,8	0,9	≥ 1				
C_β		1	0,95	0,9	0,85	0,8	0,75	0,7				

Figure 5.4: Table 21.9

Table 21.9 was also needed to find C_β . Since it is given as a function of the contact ratio ε_β , we computed that first using the following equation:

$$\varepsilon_\beta = \frac{b \cdot \tan(\beta)}{p_t} = 2.74 \quad (5.5)$$

and thus:

$$C_\beta = 0.7$$

Therefore K_V is given by:

$$K_V = 1 + \left(1 + \frac{k_e \cdot f_{pe}}{w \cdot K_I} 0.65 \text{ N/mm}\right) \cdot C_V \cdot C_\beta \cdot v = 1.11 \quad (5.6)$$

Then the load distribution factor $K_{F\alpha}$ was calculated. q_L was computed first:

$$q_L = 0.4 + \frac{f_E}{|w|}$$

With f_E calculated using the formula in table 21.10:

$$f_E = k_e \cdot f_{pe} - 8 = 20 \quad (5.7)$$

therefore:

$$q_L = 0.4 + \frac{f_E}{|w|} = 0.42 \quad (5.8)$$

Since $q_L < 0.5$ we impose $q_L = 0.5$.

Next we found ε_α with the following formula:

$$\varepsilon_\alpha = \frac{AE}{p_b} = \frac{\sqrt{r_{a2}^2 - r_{b2}^2} + \sqrt{r_{a1}^2 - r_{b1}^2} - (r_1 + r_2) \sin \alpha_t}{p_t \cos \alpha_t} = 1.631 \quad (5.9)$$

where:

$$\alpha_t = \arctan \left(\frac{\tan \alpha_n}{\cos \beta} \right) = 20.65^\circ \quad (5.10)$$

Since $q_L \leq \frac{1}{\varepsilon_\alpha}$:

$$K_{F\alpha} = 1$$

Finally we must find the longitudinal load distribution factor $K_{F\beta}$, which is set to be equal to 1 by the standard for cylindrical gears.

With all the previously calculated factors we can write the specific pitch-line force as follows:

$$w_{Ft} = 1361 \text{ N/mm}$$

Now we can calculate the tooth bending stress as:

$$\sigma_F = \frac{w_{Fi}}{m_n} Y_F \cdot Y_e \cdot Y_\beta \quad (5.11)$$

The tooth form factor Y_F was found using the equivalent number of teeth on the normal plane Z_n , along with the table A21.11, in figure 5.5 (with 21 teeth and no profile shifting):

$$Z_n = \frac{Z}{\cos^3 \beta} = 21.08 \quad (5.12)$$

$$Y_F = 2.87$$

Tab. A21.11 Zahnformfaktoren Y_F in Abhängigkeit von den Profilverchiebungsfaktoren x und den Ersatzzahnzahlen z_n bzw. z_{vn} (zusammengestellt nach DIN 3990)

z_n z_{vn}	Zahnformfaktor Y_F bei Profilverchiebungsfaktor $x =$																				
	-0,6	-0,5	-0,4	-0,3	-0,2	-0,1	0	+0,1	+0,2	+0,3	+0,4	+0,5	+0,6	+0,7	+0,8	+0,9	+1,0	+1,1	+1,2	+1,3	+1,4
7												2,84									
8											2,98	2,69	2,47								
9											2,84	2,60	2,40	2,22							
10											2,99	2,73	2,52	2,34	2,18						
11										3,15	2,87	2,65	2,46	2,30	2,16	2,05					
12										3,03	2,79	2,58	2,41	2,27	2,14	2,04					
13										2,93	2,72	2,53	2,38	2,24	2,12	2,03					
14							3,36	3,10	2,86	2,66	2,48	2,34	2,22	2,11	2,03	1,95					
15							3,25	3,01	2,79	2,60	2,44	2,31	2,20	2,10	2,02	1,95	1,89				
16							3,45	3,16	2,95	2,74	2,56	2,42	2,29	2,18	2,09	2,02	1,95	1,89			
17							3,35	3,09	2,88	2,69	2,53	2,39	2,27	2,17	2,08	2,01	1,95	1,89	1,85		
18						3,53	3,26	3,02	2,82	2,65	2,50	2,37	2,26	2,16	2,08	2,01	1,95	1,90	1,86		
19							3,72	3,44	3,20	2,96	2,78	2,61	2,47	2,35	2,24	2,15	2,07	2,01	1,95	1,90	1,87
20							3,62	3,35	3,12	2,91	2,74	2,58	2,45	2,33	2,23	2,14	2,07	2,01	1,95	1,90	1,87
21							3,53	3,28	3,07	2,87	2,70	2,55	2,43	2,32	2,22	2,14	2,06	2,01	1,95	1,91	1,87
22							3,45	3,20	3,01	2,83	2,67	2,52	2,41	2,30	2,21	2,13	2,06	2,00	1,95	1,91	1,88
23							3,64	3,38	3,15	2,96	2,80	2,64	2,50	2,39	2,29	2,20	2,12	2,06	2,00	1,95	1,91
24							3,55	3,30	3,10	2,92	2,75	2,61	2,48	2,37	2,28	2,19	2,12	2,06	2,00	1,95	1,91
25							3,73	3,45	3,25	3,05	2,88	2,72	2,58	2,46	2,36	2,27	2,19	2,12	2,05	2,00	1,95
30	3,61	3,35	3,18	3,01	2,85	2,72	2,60	2,48	2,38	2,30	2,22	2,16	2,10	2,04	2,00	1,96	1,93	1,90	1,88	1,86	1,85
40	3,15	3,00	2,86	2,75	2,63	2,54	2,45	2,37	2,30	2,24	2,18	2,13	2,08	2,04	2,01	1,97	1,95	1,93	1,91	1,90	1,89
50	2,90	2,78	2,68	2,59	2,50	2,43	2,36	2,31	2,25	2,20	2,15	2,11	2,07	2,03	2,02	1,98	1,97	1,94	1,93	1,92	1,91
60	2,75	2,65	2,57	2,50	2,42	2,37	2,32	2,25	2,22	2,17	2,13	2,10	2,08	2,04	2,02	1,99	1,98	1,96	1,94	1,94	1,93
100	2,46	2,40	2,35	2,32	2,26	2,24	2,21	2,17	2,15	2,12	2,10	2,08	2,06	2,04	2,03	2,01	2,00	1,99	1,98	1,98	1,97
200	2,27	2,24	2,21	2,19	2,17	2,15	2,14	2,12	2,10	2,10	2,08	2,07	2,05	2,04	2,04	2,02	2,02	2,01	1,98	2,00	2,00
400	2,17	2,15	2,14	2,13	2,12	2,11	2,10	2,09	2,08	2,08	2,08	2,07	2,06	2,06	2,05	2,04	2,04	2,04	2,03	2,03	2,03
∞	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07	2,07

Figure 5.5: Table A21.11

While for Y_e we use the following formula:

$$Y_e = \frac{1}{\varepsilon_\alpha} = 0.61 \quad (5.13)$$

Y_β is taken from the table below.

$$Y_\beta = 0.88$$

Table 5.1: Table for Y_β

β	0°	5°	10°	15°	20°	25°
$\geq 30^\circ$						
Y_β	1	0.96	0.92	0.88	0.84	0.79
0.75						

Now we can calculate the tooth bending stress for the pinion:

$$\sigma_{F, \text{pinion}} = 353 \text{ MPa}$$

The material of the tooth was chosen as 34CrNiMo6, and the bending stress limit was found using table A21.12 in figure 5.6:

$$\sigma_{FD, p} = 320 \text{ MPa}$$

Tab. A21.12 Zahnradwerkstoffe und deren Dauerfestigkeiten σ_{FD} und σ_{HD} (nach DIN 3990)

Lfd. Nr.	Werkstoffgruppe	Kurzzeichen nach DIN	Behandlungszustand	Härte am Kernwerkstoff	Zahnrad-Flankeneigenschaften	Dauerfestigkeitswerte Dauerwältzfestigkeit σ_{w0} N/mm ²	Dauerfestigkeitswerte Dauerschweißfestigkeit σ_{s0} N/mm ²	Zugfestigkeit R_m N/mm ²
1	Gußeisen mit Lamellengraphit nach DIN 1691	GG 20	-	170 HB	170 HB	270	50	200
2		GG 25		210 HB	210 HB	310	60	250
3		GG 35		230 HB	230 HB	360	80	350
4		GGG 40		170 HB	170 HB	360	250	400
5	Gußeisen mit Kugellengraphit nach DIN 1693	GGG 60	-	250 HB	250 HB	490	380	600
6		GGG 80		275 HB	275 HB	560	500	800
7								
8	Schwarzer Temperguß nach DIN 1692	GTS 35	-	140 HB	140 HB	360	190	800
9		GTS 65		235 HB	235 HB	490	230	1000
10	Stahlguß nach DIN 1681	GS 52	-	150 HB	150 HB	340	150	470
11		GS 60		175 HB	175 HB	420	170	520
12	allgemeine Baustähle nach DIN 17 100	St 42	-	125 HB	125 HB	290	170	450
13		St 50		150 HB	150 HB	340	190	550
14		St 60		180 HB	180 HB	400	200	650
15		St 70		208 HB	208 HB	460	220	800
16	Vergütungsstähle nach DIN 17 200	Ck 22	vergütet	140 HV	140 HV	440	170	600
17		Ck 45	normalisiert	185 HV	185 HV	590	200	800
18		Ck 60	vergütet	210 HV	210 HV	620	220	900
19		34 Cr 4	vergütet	260 HV	260 HV	650	260	900
20		37 Cr 4	vergütet	260 HV	260 HV	650	270	950
21		42 Cr Mo 4	vergütet	280 HV	280 HV	670	290	1100
22		34 CrNi Mo 6	vergütet	310 HV	310 HV	770	320	1300
23		Ck 45	umlaufgehärtet, einschließl. Zahngrund	220 HV	560 HV	1100	270	1000
24		37 Cr 4		270 HV	610 HV	1280	310	1150
25		42 Cr Mo 4		275 HV	650 HV	1360	350	1300
26	Vergütungsstähle nach DIN 17 200 nitriert	Ck 45	badnitriert	220 HV	400 HV	1100	350	1100
27		42 Cr Mo 4	badnitriert	275 HV	500 HV	1220	430	1450
28		42 Cr Mo 4	gasnitriert	275 HV	550 HV	1220	430	1450
29	Nitrierstähle nach Stahl-Eisen-Werkstoffblatt 830-60	31 Cr Mo V 9	gasnitriert	320 HV	700 HV	1400	445	1500
30	Einsatzstähle nach DIN 17 210	C 15	einsatzgehärtet	190 HV	720 HV	1600	230	900
31		16 Mn Cr 5		270 HV	720 HV	1630	460	1400
32		20 Mn Cr 5		330 HV	720 HV	1630	480	1500
33		20 Mo Cr 4		270 HV	720 HV	1630	400	1300
34		15 Cr Ni 6		310 HV	720 HV	1630	500	1600
35		18 Cr Ni 8		400 HV	740 HV	1630	500	1700
36		17 Cr Ni Mo 6		400 HV	740 HV	1630	500	1700
37	Duroplast-Schichtstoffe	Hartgewebe, grob	Bei Lauf gegen gehärtetes, feingeschliffenes Stahlrad, Öl schmierung $\leq 60^\circ\text{C}$, Umfangsgeschwindigkeit $v \leq 2\text{m/s}$			110	50	
38		Hartgewebe, fein				130	60	

Figure 5.6: Table A21.12

therefore the safety factor is:

$$S_F = \frac{\sigma_{FD, p}}{\sigma_{F, \text{pinion}}} = 0.91 \quad (5.14)$$

So far the safety factor is well below the 1.8 minimum required for infinite life. Therefore, we should either change the material of the pinion or apply a profile shift. In this section we will try the first approach: the new chosen material is 17CrNiMo6, so the bending stress limit becomes:

$$\sigma_{FD, p} = 500 \text{ MPa}$$

Thus the safety factor increases to:

$$S'_F = 1.42$$

This is still below the limit for infinite life, so we will have to apply a profile shift.

5.3 Bending Fatigue (Wheel)

The steps are the same as those for the pinion bending fatigue. A table summarizing the most important parameters is shown below:

Table 5.2: Main parameters for bending fatigue of wheel tooth

Parameter	Value	Units
k_p	188923	N
b	193	mm
w	985.6	N/mm
K_I	1.25	-
K_V	1.10	-
K_{Fa}	1	-
$K_{F\beta}$	1	-
w_{Ft}	1354	N/mm
Y_F	2.21	-
Y_α	0.61	-
Y_β	0.88	-

The tooth bending stress for the wheel is given by:

$$\sigma_{F,wheel} = 273 \text{ MPa}$$

The material chosen for the wheel is 17CrNiMo6, with a limit value of:

$$\sigma_{FD,w} = 500 \text{ MPa}$$

Thus the safety factor is:

$$S_F = \frac{\sigma_{FD,w}}{\sigma_{F,wheel}} = 1.83 \quad (5.15)$$

Since this value is greater than 1.8, this component is verified for infinite life.

5.4 Contact Fatigue (Pinion and Wheel)

The contact stress is the same for both pinion and wheel since we only have one contact between them. First, we computed the specific pitch-line force:

$$w_{Ht} = w \cdot K_I \cdot K_V \cdot K_{Ha} \cdot K_{H\beta} \quad (5.16)$$

where the specific nominal pitch-line force, the overload factor and the dynamic factor w , K_I , and K_V are the same as those computed for the pinion tooth bending strength. The spur gear load distribution factor K_{Ha} is evaluated through the following formula:

$$K_{Ha} = 1 + 2 \cdot (q_L - 0.5) \left(\frac{1}{Z_e^2} - 1 \right) = 1 \quad (5.17)$$

The longitudinal load distribution factor $K_{H\beta}$, the standards impose a value of 1 for cylindrical gears. The specific pitch-line force is given by:

$$w_{Ht} = 1352 \text{ MPa}$$

The Herzian stress is calculated through the following equation:

$$\sigma_H = \sqrt{\frac{W_{Ht}}{d_1} \cdot \frac{u+1}{u}} \cdot Z_H \cdot Z_M \cdot Z_e \quad (5.18)$$

where d_1 is the pinion pitch diameter, u is the gear ratio of the second stage i_2 . The flank form factor Z_H is computed with the following formula:

$$Z_H = \frac{1}{\cos \alpha_t} \sqrt{\frac{\cos \beta_b}{\tan \alpha_{wt}}} \quad (5.19)$$

Where β_b is given as:

$$\beta_b = \arcsin(\sin \beta \cdot \cos \alpha_n) = 14.08^\circ \quad (5.20)$$

Finally Z_H :

$$Z_H = 1.715$$

The material factor Z_M was found in table A21.13 in Figure 5.7, and by considering the material as steel we have:

$$Z_M = 272 \sqrt{N/mm^2}$$

Tab. A21.13 Werkstoffkoeffizienten Z_M nach DIN 3990

Werkstoff	Rad Kurzzeichen	Elastizitätsmodul E N/mm ²	Werkstoff	Gegenrad Kurzzeichen	Elastizitätsmodul E N/mm ²	Werkstofffaktor Z_M $\sqrt{N/mm^2}$
Stahl	St	210000	Stahl	St	210000	272
			Stahlguß	GS-60	205000	270
				GS-52	205000	270
			Gußeisen mit Kugelgraphit	GGG-60	176000	259
				GGG-40	175000	258
			Guß-Zinnbronze	G-CuSn	105000	222
			Kupfer-Zinn (Zinnbronze)	CuSn8	115000	228
			Gußeisen mit Lamellengraphit (Grauguß)	GG-25	128000	237
Stahlguß	GS-60	205000		GG-20	120000	232
			Stahlguß	GS-52	205000	268
			Gußeisen mit Kugelgraphit	GGG-60	176000	257
			Gußeisen mit Lamellengraphit (Grauguß)	GG-20	120000	231
Gußeisen mit Kugelgraphit	GGG-60	176000	Gußeisen mit Kugelgraphit	GGG-40	175000	248
			Gußeisen mit Lamellengraphit (Grauguß)	GG-20	120000	224
Gußeisen mit Lamellengraphit (Grauguß)	GG-25	128000	Gußeisen mit Lamellengraphit (Grauguß)	GG-20	120000	209
	GG-20	120000				205
Stahl	St	210000	Duroplast-Schichtstoff (Hartgewebe)		8000	80,5

Figure 5.7: Table A21.13

The contact ratio factor Z_ϵ was computed with the following equation:

$$Z_\epsilon = \sqrt{\left(\frac{4 - \epsilon_\alpha}{3} (1 - \epsilon_\beta) + \frac{\epsilon_\beta}{\epsilon_\alpha}\right) \cdot \cos(\beta_b)} = 0.551 \quad (5.21)$$

In the end we got a Hertz stress of:

$$\sigma_H = 931 \text{ MPa}$$

The allowable fatigue stress, found in Table A21.12, initially was 770 considering 34CrNiMo6, yielding a safety factor:

$$S_F = \frac{\sigma_{HD}}{\sigma_H} = 0.83 \quad (5.22)$$

which is below the 1.4 required for infinite life.

Changing the pinion material also improves the contact fatigue as the new allowable fatigue stress becomes $\sigma'_{HD} = 1630 \text{ MPa}$, therefore the new safety factor would be:

$$S'_F = 1.75$$

Thus the pinion and the wheel are verified for the contact fatigue.

5.5 Profile Shift

Then we applied a symmetric profile shifting to pinion and wheel in order to improve the safety factor for bending fatigue. Since the pinion was the most critical component, we applied a positive shift to the pinion and negative one to the wheel.

Profile shift will only affect the following two parameters, so it won't be necessary to replicate the whole calculations:

- ε_α
- Y_F

We used the following profile shift:

$$\begin{cases} x_1 = +0.2 \\ x_2 = -0.2 \end{cases} \quad (5.2)$$

Since the thickness of the pinion will increase, we can expect to increase its safety factor for tooth bending, while sacrificing that of the wheel (since its thickness will decrease).

By following all the previous steps, we get the following value for the pinion tooth bending stress:

$$\sigma_{F,p} = 314 \text{ MPa}$$

The new safety factor is:

$$S_{F,p} = \frac{500}{314} = 1.59 \quad (5.23)$$

Although this is an improvement, we still did not reach the required value of safety factor, so the pinion still isn't verified for bending fatigue.

Let's now consider the effects of profile shifting on the wheel. The new stress is:

$$\sigma_{F,w} = 289 \text{ MPa}$$

And the new safety factor:

$$S_{F,w} = 1.73$$

In the end, we can affirm that this profile shifting did not produce the expected results from the bending fatigue point of view, since now also the wheel is not verified.

Finally we analyzed the effect the contact fatigue:

The only parameter we has to changes was Z_ε , since it depends on ε_α which decreases to 1.578. This causes an increase in Z_ε to 0.582. In the end, the contact stress will rise to 988 MPa, and the new safety factor will be:

$$S_F = \frac{1630}{986} = 1.65 \quad (5.24)$$

Therefore the pinion and wheel are still verified for infinite life from the point of view of contact fatigue.

In conclusion, the profile shift wasn't effective enough in increasing the safety factor of the pinion, and so it cannot be considered the correct solution to the problem. In order to verify both components for infinite life we could:

1. Changing the material of the gears in order to find one with a higher σ_{FD} ;
2. Increase the tooth face width b , which would decrease the specific tangential load and thus decrease the stress;
3. Changing the distance between the gears, so that the overall geometry could be adapted to sustain higher loads.

If we were forced to work with the original design, the only solution would be periodical maintenance to evaluate the residual life of the gears.